Traverse CPA

KPI and Calculation Reference

Every KPI, gate and calculation in the Control Loop Performance module: what it means, the formula behind it, its thresholds, and where it appears on screen.

Item	Value
Document	kpi-documentation.docx
Scope	CPA / CPLM module: Flink calculation jobs, cplm-api, /cpm pages
Code baseline	branch feat/v25-next-release @ 97ce20f9 (SME review batch, CHG-040 … CHG-064)
Date	7 October 2026
Audience	Process / control engineers, plant reviewers, support and commissioning engineers
Method	Formulas transcribed from source code (file and line references in Appendix A); screenshots from the lab system

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1. How to read this document

Sections 2–10 explain how the engine computes each number; Section 11 walks through every page and says which number is shown where and how it is formatted; Section 2.2 specifies every window; Section 12 lists the calculation catalogue; Section 14 lists known gaps; Appendices B and C list every stored short-window and long-window value. Formulas are written as plain arithmetic; `monospace` names are the stored field or configuration key.

Term	Meaning
Window	A time slice of samples evaluated as one unit. Short windows 1m–60m (KPIs, G0–G4); long windows 4h / 12h / 24h (diagnostics); fused verdicts on 12h and 24h only.
Gate (G0–G15)	One check or detector with a status (PASS, WARN, FAIL, STRONG, …). Gates are grouped in tiers: eligibility, performance, diagnostic evidence, confirmation, fusion.
Verdict / diagnosis	The G15 fused result for a window, e.g. CONFIRMED_OSCILLATION. Its prefix is the confidence band; its suffix is the fault family.
Evidence score	The fused `confidence` (0–1) shown as 0–100. How completely the observed symptoms match a family's signature; uncalibrated, not a probability (GUM).
Severity	LOW / MEDIUM / HIGH, derived only from the evidence-score band.
Priority	High / Medium / Low = loop criticality × severity (ISA-18.2 / EEMUA 191 style).
Episode	One continuous fault of one family on one loop (Loop Events), opened and closed from LIVE verdicts.
Observability flag	A missing input or limitation on a window (NO_VP, HEALTH_NOT_CONFIGURED, NO_UPSTREAM_LINKS, NO_STEP_TEST, NO_VALID_CYCLE, …).
Good-error band	±0.5 % of the declared PV span around SP (0.5 engineering units when no PV range is declared).
PV / SP / OP / VP / MODE	Process variable, setpoint, controller output (0–100 %), valve position (optional), controller mode.

2. Where every number comes from

Every CPA KPI is computed once, in Apache Flink, from the loop's own samples (PV, SP, OP, MODE, optional VP, each with an OPC quality). The .NET service `cplm-api` stores the results, builds episodes and fleet roll-ups, and the React pages only format them. No page recomputes an engine number.

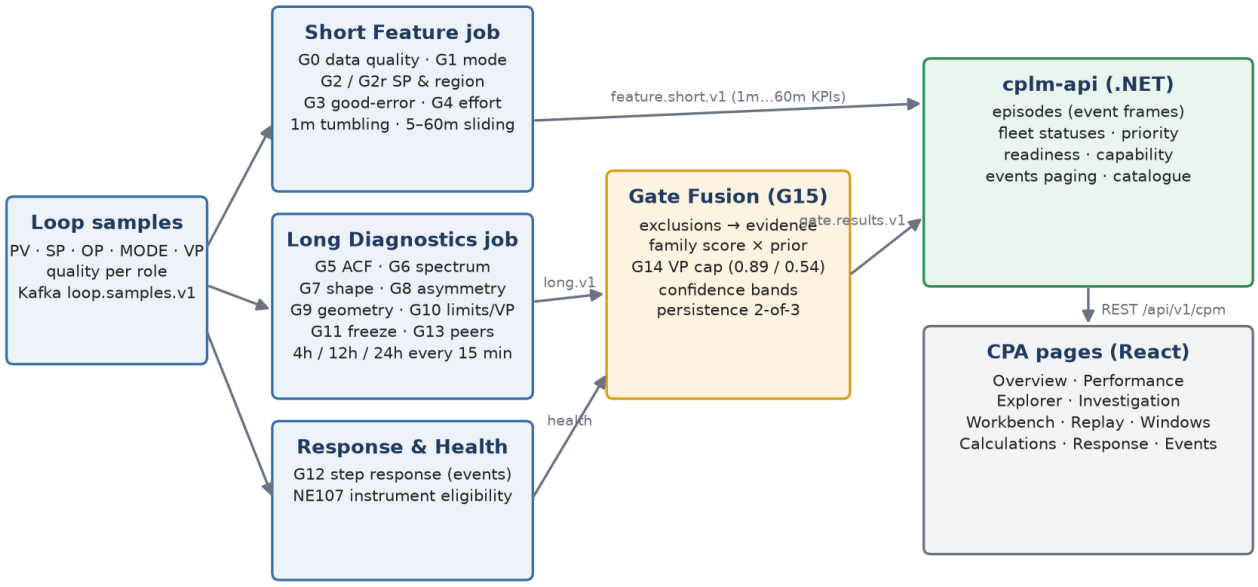


Figure 1 — Calculation pipeline (samples → Flink jobs → cplm-api → pages)

Stage	Job / component	Computes	Window and cadence	Output
Short features	`CplmShortFeatureStreamJob`	G0, G1, G2, G2r, G3, G4 and the loop KPIs (MAE, IAE, good-error time, effort…)	1m tumbling; 5m (1 min slide), 10m (2 min), 15m / 30m / 60m (5 min slide)	`feature.short.v1`
Long diagnostics	`CplmLongDiagnosticsStreamJob`	G5–G11, G13, VP tracking (G14 input), plus an aligned G0–G4 recompute	Sliding 4h / 12h / 24h, re-evaluated every 15 min; slice needs ≥ 32 samples	`feature.long.v1`
Gate fusion	`CplmGateFusionStreamJob`	Exclusions, G12 flag, G13 status, G14 confirmation, G15 diagnosis + evidence score	Fires on 12h and 24h only (every 15 min per kind)	`gate.results.v1`
System response	`CplmResponseAnalysisJob`	G12 step response: observed metrics, process model, validation	Event-driven (each qualifying SP / OP step)	`response.results.v1`
Instrument health	`CplmInstrumentContextJob`	NE107 eligibility per loop role	Per observation + 60 s freshness timer	`context.instrument-health.v1`
Episodes, fleet KPIs	cplm-api (C#)	Event frames, operating picture, priority, readiness, capability	Per fused record / per request	PostgreSQL `traverse_cplm` + REST `/api/v1/cpm`

Important: the G0–G4 values on a fused 12h / 24h verdict come from a recompute over that same 12h / 24h window, not from the 1m–60m short windows. The two can legitimately differ (the Calculations page labels each value with its window).

2.1 Sample qualification (applies to every job)

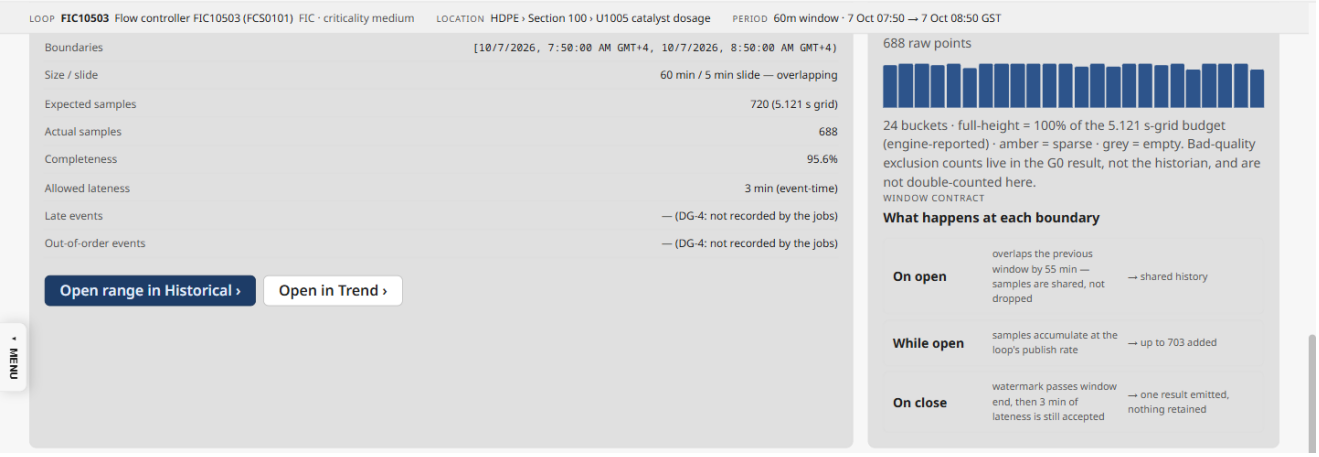
Rule	Definition
Valid sample	PV, SP and OP finite and parsed; otherwise dropped at ingest. A non-finite VP becomes "no VP" (the sample stays).
Good quality	Only the normalised token GOOD. UNCERTAIN, BAD, missing and UNKNOWN are not good. Bad samples stay in the arrays and only count in `bad_quality_pct`.
Automatic mode	AUTO, AUT, A, AUTOMATIC, NORMAL, NORM, CAS, CASC, CASCADE, RSP, DDC, SUP, SUPERVISORY (or containing AUTO / CASCADE). Manual tokens (MAN, IMAN, LO, OFF, TRACK, …) win.
Eligible VP	VP present, quality GOOD and not a held value older than 60 s (`vpMaxAgeS`).
Sample period Ts	Median observed interval between samples (5 s if fewer than 2 samples).
Statistics	σ is the population standard deviation; OP is normalised to 0–100 % of the declared OP range when one is declared.

2.2 Window specification (every window the engine evaluates)

Short windows carry the loop KPIs and gates G0–G4; long windows carry the diagnostics G5–G11 and G13 (and an aligned G0–G4 recompute); only the 12h and 24h long windows produce a fused verdict (G12, G14, G15). All windows are event-time, half-open [start, end).

Window	Tier	Type	Size / slide (new result every)	Allowed lateness	Minimum data	Gates	Fused verdict
1m	short	tumbling	1 min / —	30 s	≥ 10 samples and G0 not FAIL	G0–G4	no
5m	short	sliding	5 min / 1 min	60 s	same	G0–G4	no
10m	short	sliding	10 min / 2 min	90 s	same	G0–G4	no
15m	short	sliding	15 min / 5 min	2 min	same	G0–G4	no
30m	short	sliding	30 min / 5 min	2 min	same	G0–G4	no
60m	short	sliding	60 min / 5 min	3 min	same	G0–G4	no
4h	long	rolling buffer	4 h / 15 min	n/a (recomputed)	slice emitted only with ≥ 32 samples	G5–G11, G13 (+ G0–G4)	no
12h	long	rolling buffer	12 h / 15 min	n/a (recomputed)	same	G5–G11, G13 (+ G0–G4)	yes → G12, G14, G15
24h	long	rolling buffer	24 h / 15 min	n/a (recomputed)	same	G5–G11, G13 (+ G0–G4)	yes → G12, G14, G15

Rule	Definition
Short window declined	`sufficient_data = false` when G0 = FAIL or n < 10: metrics are null, G1–G4 = NOT_EVALUATED; shown as DECLINED
Long window unqualified	`long_metrics_qualified = false` when the aligned G0 failed; shown as UNQUALIFIED / Insufficient data
Expected samples	round(window seconds / grid); grid = declared acquisition grid, else the median observed interval (5 s fallback)
Overlap	sliding windows overlap the previous one (e.g. 60m / 5 min shares 55 min); each emission is a separate stored row
Long buffer	24 h + 10 min of samples per loop, re-evaluated on a 15-min event-time timer; every 15 min a new 4h, 12h and 24h row
Fused verdict	12h and 24h long rows only; its G0–G4 come from the same 12h / 24h window, not from a short window



Window Inspector — the contract of one 60m window (boundaries, expected vs actual samples, lateness) and its raw sample density



Window Inspector — one KPI (MAE) across every short window size: latest, median and range of the last 12, and windows declined

3. Gate tiers and status vocabulary

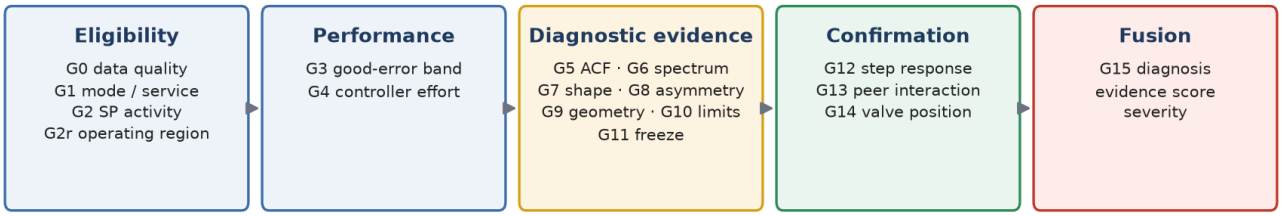


Figure 2 — The sixteen gates, by the question they answer

Mark on screen	Engine status	Meaning	Colour
✓ OK	PASS (eligibility, performance, confirmation tier)	Check passed on full inputs	Green
✓ OK on available data	PASS while one of the gate's inputs is missing (CHG-064)	Passed, but judged without e.g. VP, a health source, peers or a step test	Grey
! Attention	WARN / REVIEW (check tiers)	Check needs attention	Amber
✕ Failed	FAIL (performance / confirmation)	Check failed	Red
⊘ Not usable	FAIL on an eligibility gate, EXCLUDED_*	Window cannot be judged	Grey
◯ Not seen	PASS on a diagnostic gate (G5–G11)	Symptom not seen by this method	Grey
◐ Seen	WARN on a diagnostic gate	Symptom seen	Amber
● Strong	STRONG / FAIL on a diagnostic gate	Strong symptom evidence	Amber
△ Suspected / ▲ Confirmed	SUSPECTED / CONFIRMED (G14, G15)	Condition suspected / confirmed	Amber / Red
— No verdict	NOT_EVALUATED, PENDING, N/A, INSUFFICIENT_*	Not evaluated	Grey

ISA-101: colour is never the only cue (shape + word), and red is reserved for a failed performance check or a confirmed condition. Marks come from `gateStatus.ts`; the fleet "Needs attention" counts use the older grouping `reviewSeverity` (FAIL / STRONG / EXCLUDED / CONFIRMED = bad; WARN / REVIEW / SUSPECTED = warn).

4. Loop performance KPIs and gates G0–G4 (short windows)

Computed in `ShortFeatureCalculator` for every 1m … 60m window. If G0 fails or the window has fewer than 10 samples, the window is marked insufficient: its metrics are null and G1–G4 read NOT_EVALUATED.

Core loop KPI formulas

```
e(i)           = SP(i) − PV(i)                        control error, PV units
MAE            = mean |e|          RMSE = sqrt(mean e²)          (sample means)
IAE            = Σ |e(i)|·Δt(i)    ISE  = Σ e(i)²·Δt(i)          Δt(i) = t(i+1) − t(i)
ITAE           = Σ (t(i) − t_start)·|e(i)|·Δt(i)
band           = 0.005 × (PV_max − PV_min)            0.5 EU if no PV range declared
good_error_pct = count(|e| ≤ band) / n                "% in band", "Good-error time"
auto_pct       = count(mode is automatic) / n         "Time in auto" (per sample)
saturation_pct = count(OP ≤ 5 % or OP ≥ 95 %) / n
OCE            = auto_pct × good_error_pct × (1 − saturation_pct)
effort_ratio   = σ(OP) / σ(PV)
activity       = (σ(OP)/100) / (σ(PV)/(PV_max − PV_min))   (else effort_ratio)
```

Gate	Calculated values	Status rule (default thresholds)	Configuration keys
G0 Data quality	`completeness` = n / round(window / grid); `bad_quality_pct` = non-GOOD / n; duplicates; gaps (Δt > 1.5·Ts)	FAIL if bad ≥ 0.50; PASS if completeness ≥ 0.98 and bad < 0.05 and no duplicates; WARN if completeness ≥ 0.95; else FAIL	`G0.coverage_min` 0.95, `G0.bad_fraction_max` 0.05 (0.98 and 0.50 are fixed)
G1 Mode / service	`auto_pct`, `mode_changes_per_h`	PASS ≥ 0.90; WARN ≥ 0.70; else EXCLUDED (window excluded from diagnosis)	`G1.auto_fraction_min` 0.70 (0.90 fixed)
G2 SP activity	sp_range = max SP − min SP; band = 1 % of PV span (5 % for averaging loops)	PASS if sp_range < band, else WARN (never FAIL)	`G2.stability_band_span_fraction`, `G2.stability_band_pv_eu`
G2r Operating region	share of samples with PV or OP outside the commissioned region	PASS ≤ 0.05; WARN ≤ 0.20; else FAIL (→ EXCLUDED_OPERATING_REGION)	region bounds `G2r.*` (0.05 / 0.20 fixed)
G3 Base performance	good_error_pct, MAE, RMSE, IAE, ISE, ITAE, OCE	PASS if good_error_pct ≥ 0.50 (0.30 for averaging loops), else WARN	`G3.error_band_span_fraction` 0.005; `G3.objective_compliance_min` (×100)
G4 Controller effort	`activity_ratio_norm` (or `effort_ratio`), OP travel/day, reversals/h, saturation, freeze run	STRONG if activity > 8; WARN if > 3; else PASS	`G4.activity_warn_ratio` 3, `G4.activity_high_ratio` 8

5. Diagnostic evidence G5–G11 and G13 (long windows)

Computed in `LongDiagnosticsCalculator` on 4h / 12h / 24h windows every 15 minutes. The shape signal x is PV for integrating loops (level, gas pressure) and OP otherwise; it is detrended before the oscillation methods.

Long-window formulas

```
ACF        ρ(k) = [Σ x(i)·x(i+k)/(n−k)] / γ0,  γ0 = Σ x²/n,  k = 1 … min(n/2, 2000)
period     P = median of 2·(z(j+1) − z(j))·Ts over ACF zero crossings  (acf_period_s)
regularity r = min(1, P / (3·σ(periods)))                              (acf_regularity)
spectrum   Welch PSD, Hann, 50 % overlap;  peak_ratio = peak-lobe energy / total
harmonics  HAR = (A2 + A3 + A5) / A1       HER = (A2² + A3² + A5²) / A1²
shape      cycle score = SSE_sine / (SSE_sine + SSE_tri);  triangularity = mean
asymmetry  ϕ±(k) = xcorr of standardised OP, PV
           oddness = Σ|ϕ+ − ϕ−| / (Σ|ϕ+ − ϕ−| + Σ|ϕ+ + ϕ−|)
geometry   area_norm = mean |shoelace area per cycle| / (ΔPV_range · ΔOP_range)
limits     saturation = count(OP ≤ 5 or OP ≥ 95 pp) / n;  dwell = longest run at limit
freeze     freeze_index_s = longest run of |ΔPV| ≤ 1e-9, × Ts
           material freeze ⇔ ≥ 60 s and ≥ 10 % of the window
```

Gate	Question	Status rule (defaults)	Notes
G5 Oscillation (ACF)	Is there a regular cycle?	WARN when any ACF period exists, else PASS	Period validation (band τmin–τmax, regularity ≥ 0.10, ≥ 8 samples/period) is separate and feeds G7 / G9 / fusion
G6 Spectrum	Is one frequency dominant?	WARN if peak_ratio > 0.5, else PASS	Fallback period source (FFT, harmonic fold k = 2…5)
G7 Shape	Triangular (stick-slip) rather than sine?	NOT_EVALUATED without a valid period; STRONG if triangularity > 0.8; else PASS	`G7.shape_score_high` 0.8
G8 Asymmetry	Is the PV/OP lag asymmetric?	STRONG if oddness > 0.7; else PASS	Needs ≥ 202 samples; `G8.asymmetry_display_threshold` 0.7
G9 Phase geometry	Does the PV–OP loop have stiction corners?	NOT_EVALUATED without valid cycles, area above band, effort or OP travel below floor; PASS if area below band; STRONG if area > 0.30; else PASS	Area band 0.05–1.5 (2.0 slow / integrating); role per class
G10 Limits / valve	Is OP stuck at or cycling against a limit?	WARN if occupancy ≥ 0.05, or dwell ≥ 30 samples × Ts, or limit cycling (≥ 3 exits, ≥ 2 %)	Also computes VP tracking: |OP − VP| mean / P95, non-following dwell
G11 Sensor freeze	Is the measurement frozen?	WARN MATERIAL_FREEZE (→ EXCLUDED_SENSOR) or WARN FREEZE_EPISODE; else PASS	`G11.freeze_dwell_s` 60, tolerance `G11.freeze_tolerance_pv_eu`
G13 Interaction	Does a peer loop move with this one?	WARN if a peer has coherence ≥ 0.6 and |xcorr| ≥ 0.5 within ±600 s (support ≥ 1800 s); PASS if none; NOT_EVALUATED without peers	Up to 12 peers; direction leads / follows / simultaneous

6. G12 System response (step response)

The Response Engine watches each loop in automatic mode with good quality for a setpoint or output step, captures the response, computes observed metrics and fits a process model. On the fused verdict, G12 is only a registry flag (PASS when an approved step test is recorded, else NOT_EVALUATED with flag NO_STEP_TEST).

System response formulas

```
trigger T  = max(1.0 EU, 0.01 × 100, 4 × 1.4826 × MAD(pre-window))   pre-window 120 s
step       |x − median(pre)| ≥ T and the new level holds within T/2 for 30 s
capture    ≥ 180 s, ≤ 4 h; censored on a gap > 60 s, mode change or newer step
progress   p(t) = (y(t) − y0) / (yss − y0)
           t10, t50, t63, t90 = first crossing held ≥ 5 s
rise time  t90 − t10      overshoot % = 100·(max p − 1)    decay = (p2 − 1)/(p1 − 1)
settling   PV stays within max(0.1 EU, 2 % of |Δy|, 0.05) of yss
servo gain Δy / (r1 − r0)     steady error = r1 − yss     IAE / ISE / ITAE
model      FOPDT (LIC: IPDT): gain Kp, dead time θ, time constant τ; AICc
validate   improvement = 1 − RMSE_free / RMSE_baseline (on the next event)
           ≤ 0 → REJECTED;  RMSE or bias above noise → CANDIDATE;  else VALIDATED
```

Output	Shown as	Unavailable when
baseline_pv / endpoint_pv	Start and settled PV (robust medians)	Baseline not stable, or no stable tail
t10 / t50 / t63 / t90, rise time	Response timing (s)	Censored capture, or response < 0.5 EU
settle_response / settle_target	Settling time (s)	Never settled inside the capture
overshoot / undershoot / decay ratio	% and ratio	No qualified peak
process_gain Kp, dead_time θ, time_constant τ	Process model card (with standard errors)	No excitation, fit failed, budget (20 s) exceeded
model_state	CANDIDATE / VALIDATED / REJECTED	First event is always WAITING (needs an independent event)

Not produced in this build: stability margins and SIMC tuning advice (the job has no controller parameters), and set-point steps on integrating (LIC) loops (marked UNSUPPORTED).

7. G14 valve confirmation and G15 fused diagnosis

7.1 Evaluation order

- Insufficient short window → `INSUFFICIENT_DATA`.
- Blocking exclusions, first match wins: G0 FAIL → `EXCLUDED_DATA_QUALITY`; G1 EXCLUDED → `EXCLUDED_MODE`; G2r FAIL → `EXCLUDED_OPERATING_REGION`; PV health INELIGIBLE → `EXCLUDED_INSTRUMENT_HEALTH`; material freeze → `EXCLUDED_SENSOR`.

- Context gates G12, G13 and the observability flags (NO_VP, HEALTH_NOT_CONFIGURED, NO_UPSTREAM_LINKS, NO_STEP_TEST, …).
- Detector scores → evidence predicates → qualified families → selected family → confidence → band → persistence.

7.2 Detector scores and family scores

Score	Formula (clamped 0–1)	Source gate
oscillation	acf_regularity	G5
fft	fft_peak_ratio	G6
effort	effort_ratio / 8	G4
stiction	triangularity	G7
horch (asymmetry)	horch_oddness	G8
geometry	corner_score_qualified (if a valid period)	G9

Fusion arithmetic

```
family_stiction    = max(stiction, horch) × prior_stiction        (if stiction qualified)
family_oscillation = max(oscillation, fft) × prior_oscillation    (if oscillation qualified)
family_effort      = effort × prior_effort                        (if effort qualified)
family_geometry    = corner_score_qualified × prior_geometry      (if geometry selectable)
selected           = argmax over qualified families
                     (ties: stiction, oscillation, effort, geometry)
confidence         = min(family score, G14 cap)   (cap on valve claims only)
qualified          ⇔ score ≥ 0.55  or a related gate is STRONG / WARN
```

Stiction additionally needs oscillation evidence, a shape cue (G7 strong, stiction ≥ 0.55 or a qualified corner ≥ 0.5) and actuator stress, with at least two independent non-geometry evidence groups. A detected peer interaction with oscillation but no actuator stress blocks stiction (DISTURBANCE_CONTEXT → leading cause "external disturbance").

7.3 G14 valve-position confirmation

State	Condition	Cap on a valve claim
CONFIRMED	VP does not follow OP: |OP − VP| ≥ 5 pp for ≥ 60 s (non-following dwell)	1.00
CONTRADICTED	≥ 30 eligible VP samples, P95 |OP − VP| < 5 pp, dwell < 60 s (valve tracks)	0.54
NOT_CONFIRMED	No VP, or no positive evidence	0.89
NOT_APPLICABLE	Selected family is oscillation or effort	—

7.4 Confidence bands, severity and the evidence score

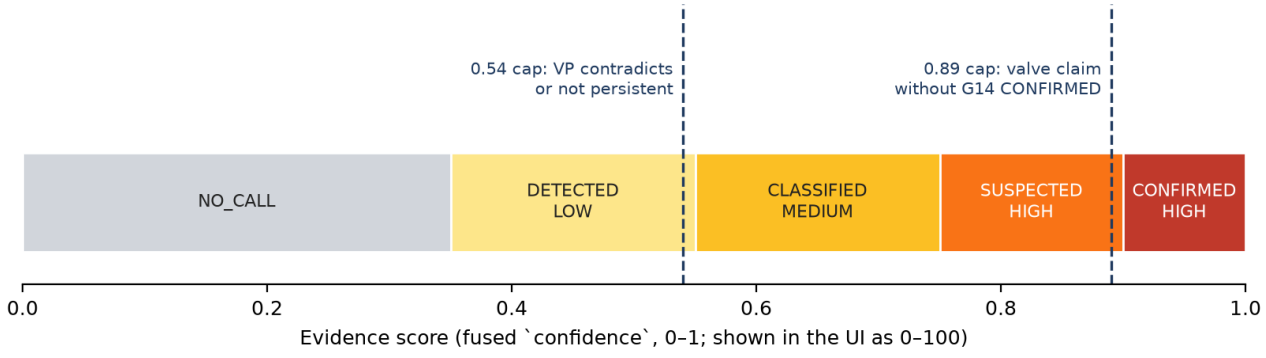


Figure 3 — G15 bands: diagnosis prefix and severity follow from the evidence score

Evidence score c	Diagnosis	G15 status	Severity
c < 0.35	NO_CALL	INSUFFICIENT_EVIDENCE	LOW
0.35 ≤ c < 0.55	DETECTED_<family>	SUSPECTED	LOW
0.55 ≤ c < 0.75	CLASSIFIED_<family>	SUSPECTED	MEDIUM
0.75 ≤ c < 0.90	SUSPECTED_<family>	SUSPECTED	HIGH
c ≥ 0.90	CONFIRMED_<family> (valve family only with G14 CONFIRMED, else SUSPECTED at 0.89)	CONFIRMED	HIGH

Families: stiction / geometry → FINAL_ELEMENT_NONLINEARITY; oscillation → OSCILLATION; effort → EXCESSIVE_EFFORT. The UI shows c × 100 as "evidence score N" (never "% confidence"): it is an uncalibrated match score, not a probability. Severity depends only on the band.

7.5 Persistence

```
need = 2, window = 3                          (persistenceMinAgree, persistenceWindows)
agree = 1 + last 2 fused results (same window kind) with the same elevated family
if agree < need and the result is SUSPECTED / CONFIRMED:
    diagnosis → DETECTED_<family>, severity LOW, confidence ≤ 0.54
    flag PERSISTENCE_CAP
```

7.6 Observations and possible causes (v2 verdict)

Observation	Present when	Score	Role
OSCILLATION	oscillation qualified	max(osc, fft)	Primary
SHAPE_TRIANGULARITY	G7 STRONG/WARN or stiction ≥ 0.55	stiction	Primary
LAG_ASYMMETRY	G8 STRONG/WARN or horch ≥ 0.55	horch	Supporting
PHASE_GEOMETRY	geometry evidence valid	geometry	Per class
ACTUATOR_STRESS	G4 / G10 STRONG/WARN, effort ≥ 2× floor, OP travel ≥ floor, cycling, ≥ 3 reversals	effort	Supporting
LIMIT_OCCUPANCY	G10 STRONG/WARN or open limit condition	min(1, 2 × saturation)	Supporting
FREEZE	material freeze or freeze episodes	1.0 / 0.5	Primary
PEER_INTERACTION	G13 interaction detected	max coherence	Supporting

Leading cause: stiction/geometry → Valve stiction; oscillation → Controller tuning (or External disturbance when DISTURBANCE_CONTEXT); effort → Excessive actuator effort. ACTUATOR_LIMIT and SENSOR are scored and shown, but never lead. The pages list Observed / Possible cause / Ruled out / No evidence from this structure.

8. Episodes (Loop Events)

An episode (event frame) is one continuous fault of one family on one loop and window kind. Built in cplm-api from LIVE fused results; replays never touch episodes.

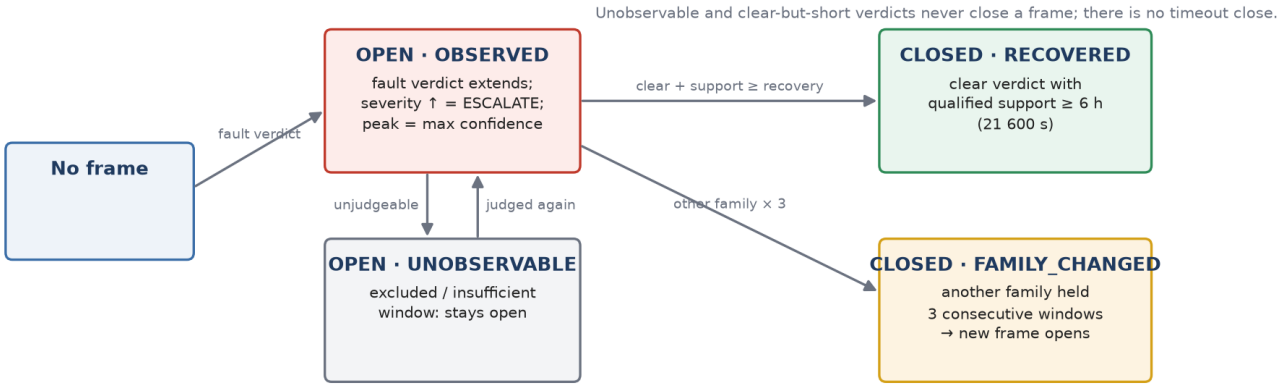


Figure 4 — Episode lifecycle

Field	Rule
Open	First fault verdict (any diagnosis except NO_CALL, INSUFFICIENT_*, EXCLUDED_*); opened_at = window end
Peak / last	peak = highest evidence score seen for the episode's family; last = newest verdict
Escalate	Severity band rises (LOW → MEDIUM → HIGH)
qualified_support_ms	Union of judged time (overlapping 15-min re-evaluations counted once)
window_count	+1 only for a non-overlapping window (≈ once per 12 h / 24 h)
Unobservable	Excluded / insufficient window: availability = UNOBSERVABLE, the condition stays OPEN
Close RECOVERED	Clear verdict with qualified support ≥ 21 600 s (6 h, `G15.recovery_support_s`)
Close FAMILY_CHANGED	Another family held for 3 consecutive windows; a new episode opens
Shelving	Separate lifecycle (ISA-18.2): SHELVED until `shelve_until`, then EXPIRED by a 60 s sweeper; acknowledging never ends a shelf

9. Instrument health (NAMUR NE107)

Condition (first match wins)	Eligibility	Shown as
No device bound	NOT_CONFIGURED	Not configured · not checked
No health interval yet	UNKNOWN	Unknown · not checked
Collection STALE / OFFLINE / UNKNOWN	LIMITED	Limited
F (failure) affecting the signal, or measurement INVALID	INELIGIBLE → window EXCLUDED_INSTRUMENT_HEALTH	Fault
C (check) affecting the signal	INELIGIBLE	Fault
S (out of spec) affecting the signal; M with degraded measurement; impact unknown	LIMITED → flag HEALTH_LIMITED	Limited
M only, flags on another variable, or no active diagnostic	ELIGIBLE	OK

Freshness window = max(3 × scan interval (30 s), 120 s). Health never raises or lowers the evidence score; it only excludes a window (INELIGIBLE PV) or adds a flag. No health source means a sensor fault is not checked.

10. Default thresholds per loop class

Resolution order: class pack → `loop-dynamics-profiles.yaml` → governed bundle (`factory-layer.json`, edited on Gate Configuration). Values below are the factory defaults.

Setting	FIC	PIC liquid	PIC gas	PIC vapour	LIC	TIC	Unknown
Dynamic class	fast self-reg	fast self-reg	integrating	slow self-reg	integrating	slow self-reg	fast self-reg
Objective	tight	tight	averaging	tight	averaging	tight	tight
Period band τmin–τmax (s)	10–900	5–1 800	120–14 400	300–28 800	120–14 400	300–28 800	5–86 400
Min valid cycles	5	5	3	3	3	3	5
G3 PASS min good-error	0.50	0.50	0.30	0.50	0.30	0.50	0.50
G2 SP band (fraction of span)	0.01	0.01	0.05	0.01	0.05	0.01	0.01
G9 corner angle (°)	65	60	50	45	50	45	65
G9 area band	0.05–1.5	0.05–1.5	0.05–2.0	0.05–2.0	0.05–2.0	0.05–2.0	0.05–1.5
G9 role / geometry prior	supporting / 0.5	supporting / 0.5	display / 0.3	primary / 1.0	display / 0.3	primary / 1.0	display / 0 (off)
Min OP travel for G9 (pp)	2.0	2.0	1.5	1.0	1.5	1.0	2.0
Shape signal	OP	OP	PV	OP	PV	OP	OP

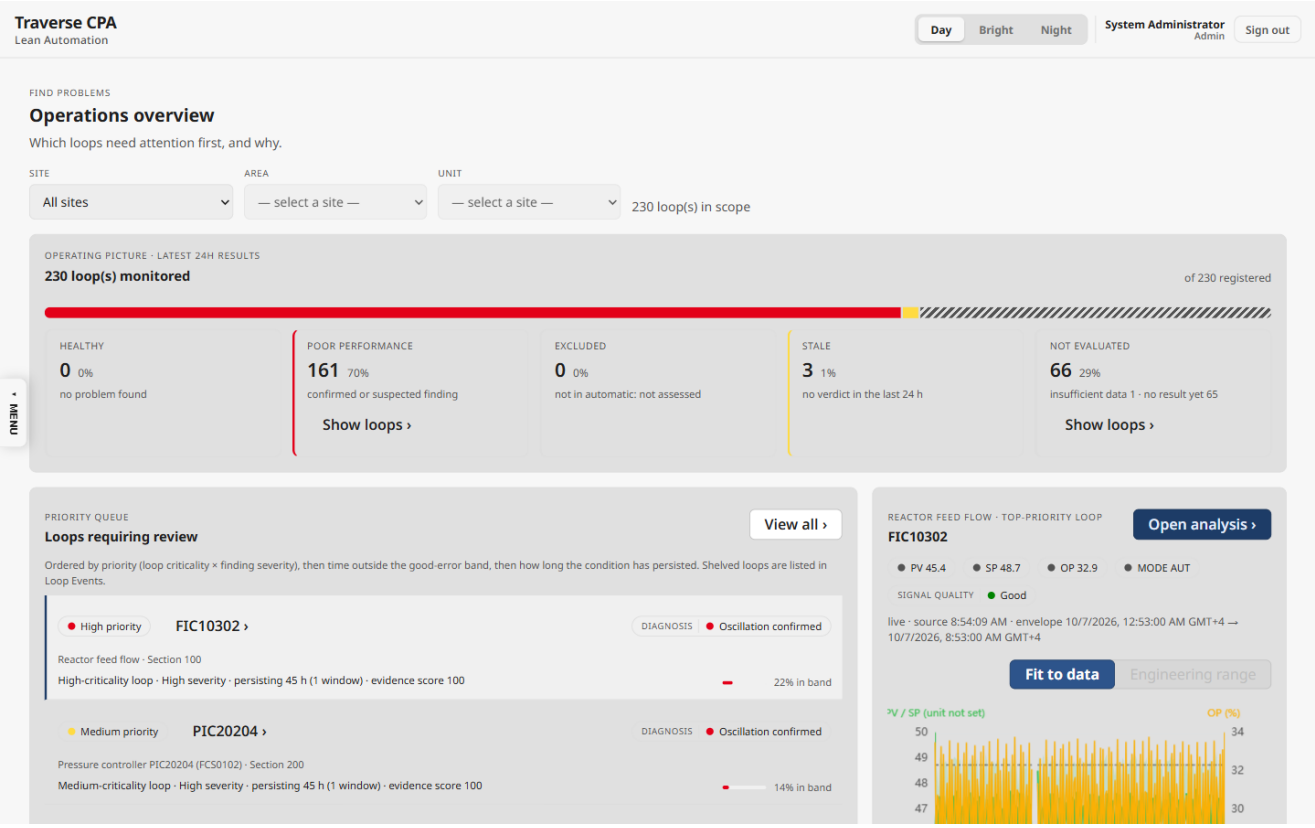
All classes: G4 WARN > 3 / STRONG > 8; G10 limit tolerance 5 pp, occupancy 0.05; VP non-following 5 pp for 60 s; persistence 2 of 3; recovery support 21 600 s; good-error band 0.5 % of PV span.

11. Page-by-page KPI reference

Screenshots from the lab (7 October 2026, day theme, 1440 px). Values on the screenshots are lab data and illustrate the layout only.

11.1 Operations overview (/cpm)

Which loops need attention first, and why. Fixed to the 24h window kind; server cache 15 s, UI refresh 60 s.



Overview — operating picture, priority queue, loop focus

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Loops in scope / monitored	in scope = registered loops in the plant scope; monitored = `monitoring.enabled = true`	integers	`fleet/summary`
Operating picture (5 tiles + bar)	each monitored loop in exactly one state, from its newest LIVE real verdict: none → Not evaluated (insufficient / no result); window ended > window size ago → Stale; EXCLUDED_* → Excluded; NORMAL/NONE/HEALTHY → Healthy; CONFIRMED/SUSPECTED/DETECTED/CLASSIFIED_* → Poor performance; else unclassified (counted in Not evaluated)	count + round(count / monitored × 100) %; Healthy green, Poor red, Stale amber, others grey	`fleet/summary` → `statuses`
Priority queue (6 rows)	current findings only (verdict within one window size, finding diagnosis), minus actively shelved episodes; level = criticality × severity matrix (Figure 5)	High red · Medium amber · Low grey	`fleet/rankings?orderBy=priority`
Queue order	level → lowest good-error % → oldest open episode → highest evidence score → loop id	—	same
Reason line	"{criticality} loop · {severity} severity · persisting {age} ({n} windows) · evidence score N"	N = round(confidence × 100)	same
% in band bar	good_error_pct (0–1) × 100	≥ 80 green, ≥ 50 amber, < 50 red	same
N shelved ›	current findings whose open episode is actively shelved	shown when > 0	same
Loop focus (PV/SP/OP/MODE, signal quality)	live Sparkplug values; quality GOOD/UNCERTAIN/BAD (OPC ≥ 192 good, ≥ 64 uncertain); no update for > 10 min → "Uncertain · last value {age}"	1 dp	MQTT live plane
Event trail	newest 3 episodes, opened_at desc, active shelves hidden	date-time + diagnosis	`events?sort=recent&limit=3`

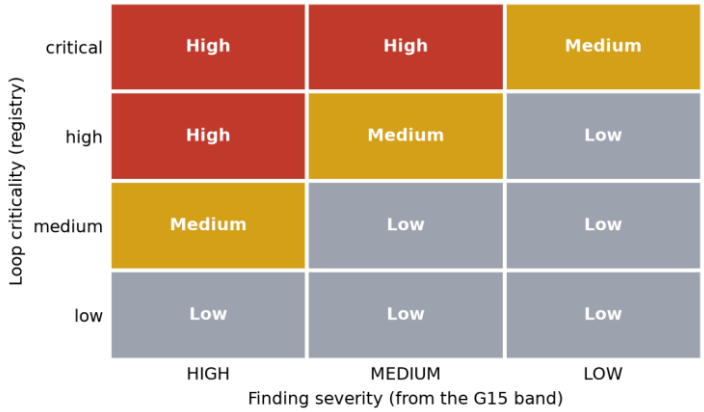
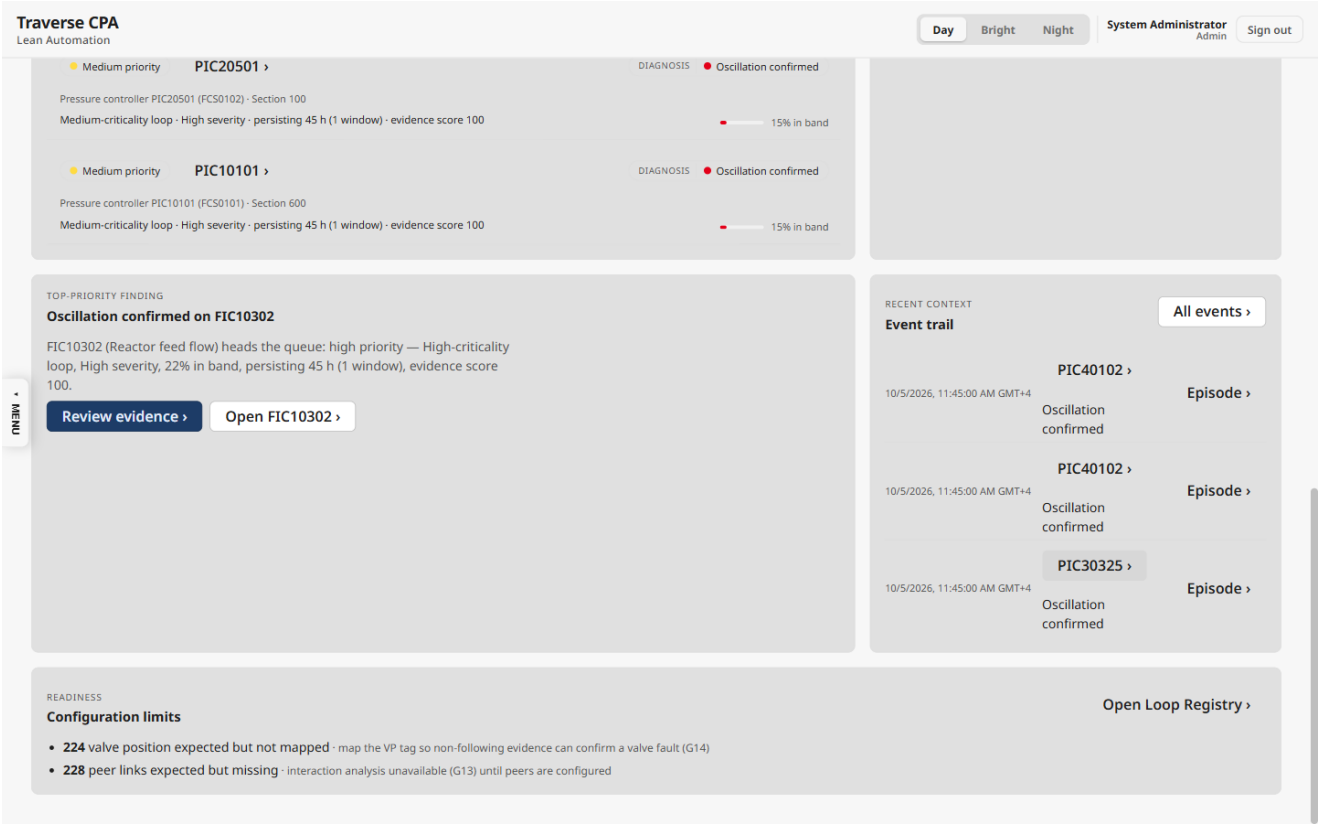


Figure 5 — Priority level = loop criticality × finding severity (ISA-18.2 / EEMUA 191 style)

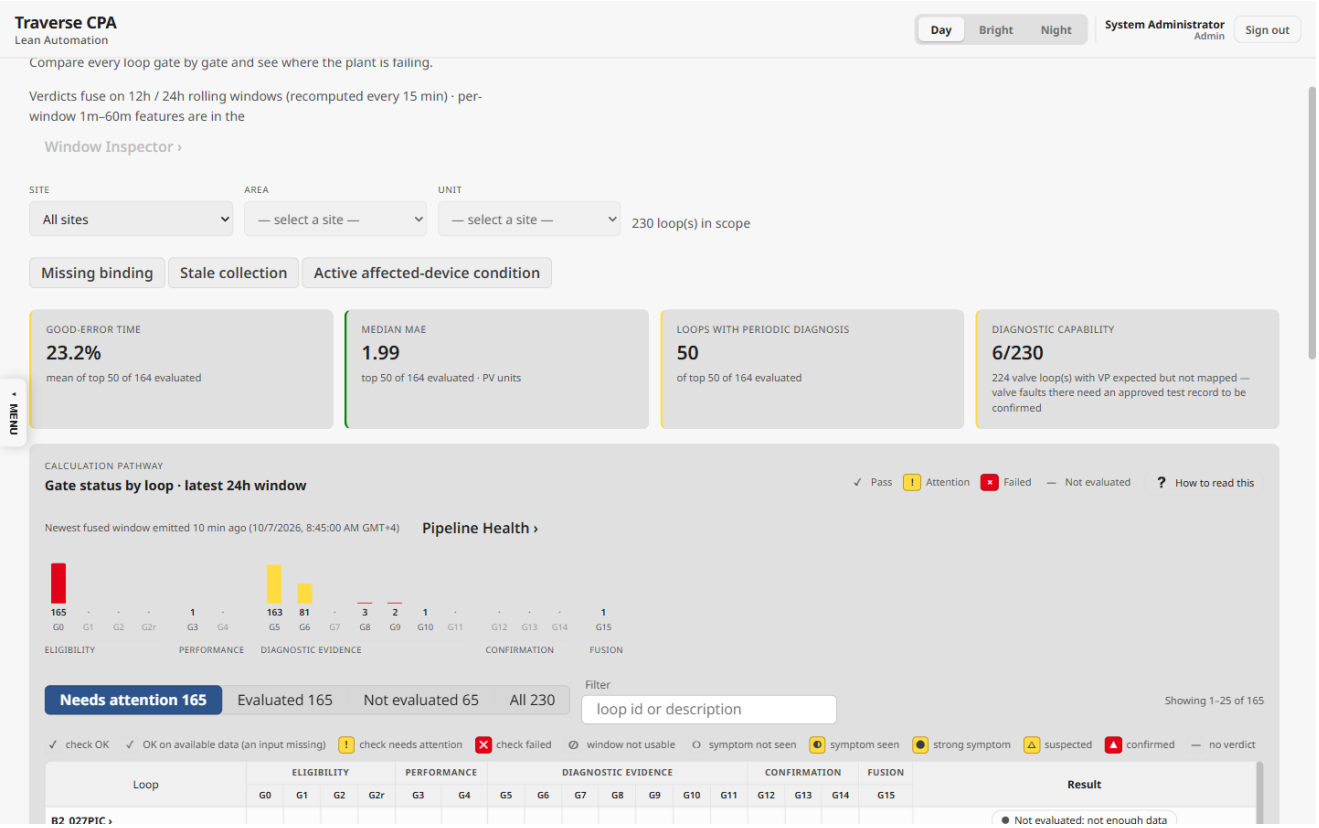


Overview — top-priority finding, event trail and readiness strip

KPI on screen	Logic / formula	Display and thresholds	Endpoint
VP expected but not mapped	registry loops with no vp tag and VP applicability EXPECTED (declared, else command path: OP_TO_VFD / OP_TO_PV → not applicable)	count + remedy text	`fleet/summary` → `capability`
Peer links expected but missing	no configured peer links and peer applicability EXPECTED	count	same
Plant facts	VP not installed / not applicable; peers not applicable — counted apart, never as gaps	grey sentence	same

11.2 Performance (/cpm/performance)

Compare every loop gate by gate and see where the plant is failing. Window toggle 12h / 24h.

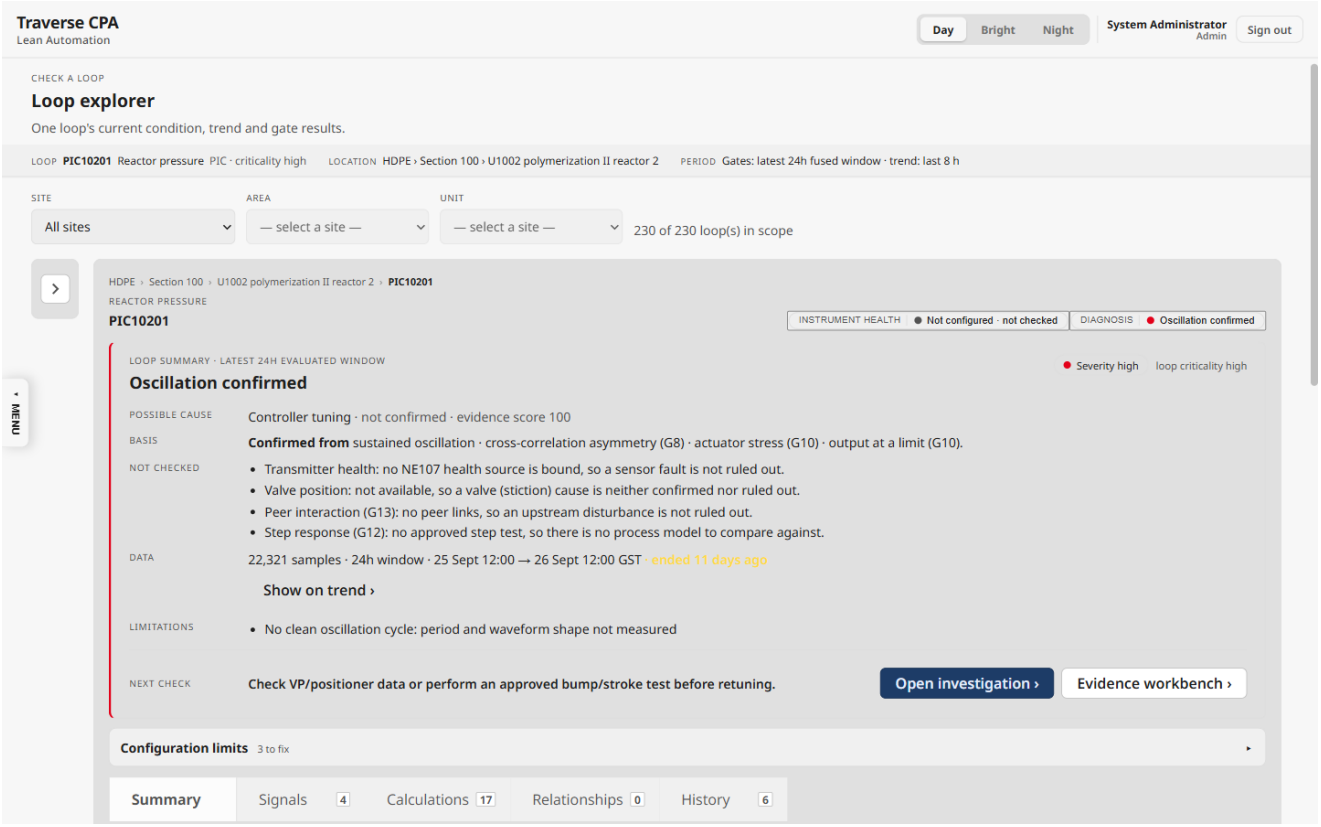


Performance — KPI tiles, gate roll-up and gate matrix

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Good-error time	100 × mean(good_error_pct) over the evaluated loops of the top-50 ranking	1 dp %; green ≥ 80, else amber	`fleet/rankings`
Median MAE	median of MAE (> 0) over the same loops	2 dp, PV units (mixed units across loops)	same
Loops with periodic diagnosis	evaluated loops whose diagnosis contains OSCILLATION or FINAL_ELEMENT	amber if > 0	same
Diagnostic capability N/M	M = loops with a valve (registered − VP not applicable); N = M − VP expected-unmapped − VP not installed	amber if any expected-unmapped	`fleet/summary`
Gate roll-up bars	per gate: loops whose status is bad (FAIL, STRONG, EXCLUDED, CONFIRMED) or warn (WARN, REVIEW, SUSPECTED)	stacked red / amber	`fleet/heatmap`
Needs attention / Evaluated / Not evaluated / All	attention = any gate bad or warn; evaluated = diagnosis ≠ NOT_EVALUATED	counts	same
Gate matrix cell	gate mark from status (Section 3)	glyph + tone	same
Freshness	now − newest fused window end	stale styling after 3 h	same
Attention list	top 12 by evidence score, or by good-error % ascending	"evidence score N" / "N% in band"	`fleet/rankings`

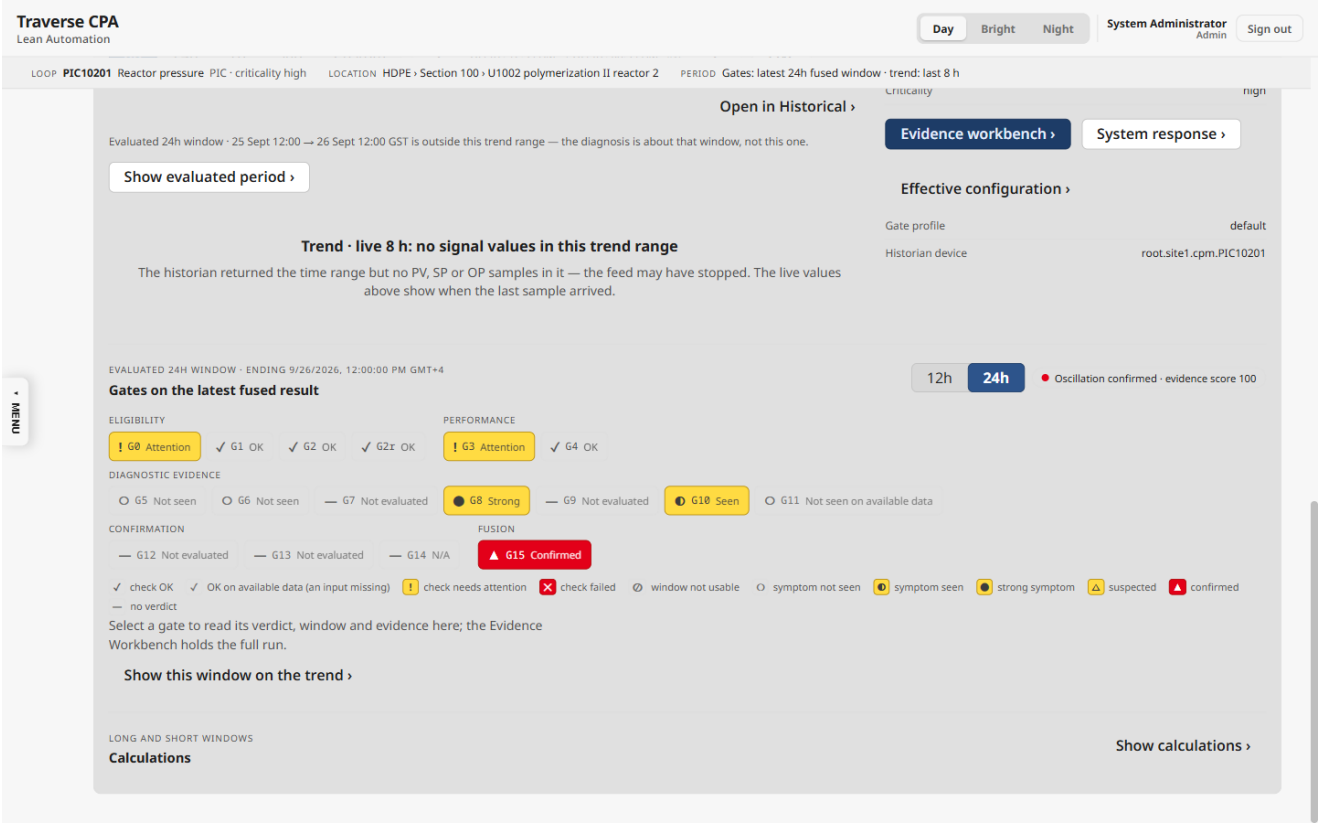
11.3 Loop Explorer (/cpm/explorer)

One loop's current condition, trend and gate results; the Loop summary card heads every tab.



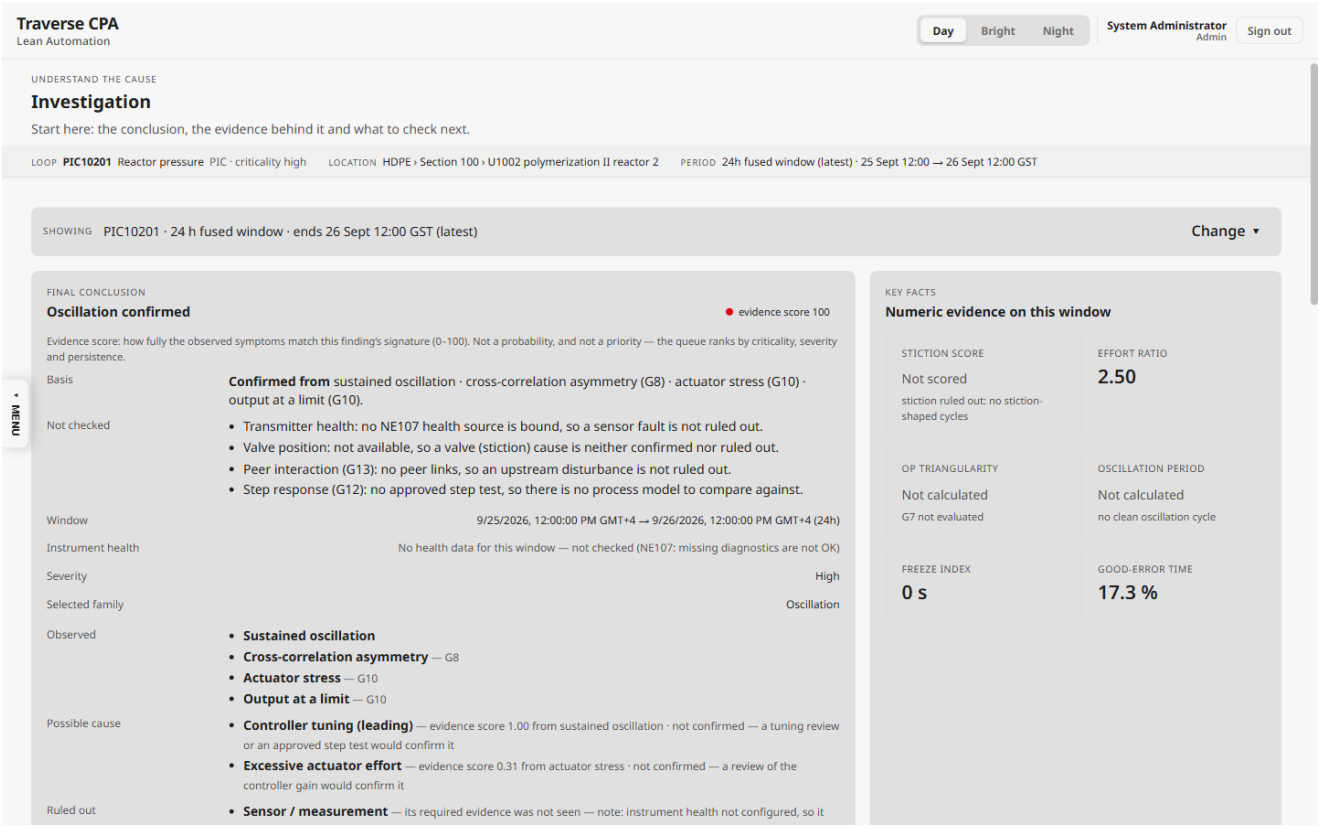
Loop Explorer — facet pills and Loop summary card (Basis / Not checked)

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Instrument health pill	no data → Unknown; no role configured → Not configured; any INELIGIBLE → Fault; any LIMITED → Limited; else OK	"· not checked" when not configured / unknown; Fault red, Limited amber, OK green	`instrument-health`
Diagnosis pill / Condition	latest fused window's diagnosis → words (e.g. CONFIRMED_OSCILLATION → "Oscillation confirmed")	tone by band	`gates/latest`
Severity	engine band LOW / MEDIUM / HIGH (finding only) + loop criticality	pill	same
Possible cause · evidence score	leading hypothesis label; score = round(confidence × 100); cap text when at 54 (not persistent) or 89 (no VP)	"evidence score 89 (capped at 89: …)"	same
Basis	"Confirmed from" (CONFIRMED) / "Based on": present observations, citing only gates that fired	sentence	same
Not checked	inputs the verdict was judged without (health, VP, peers G13, step test G12), skipping those that do not apply to the loop	bullets	same + health + registry
Data	sample count · window · "ended {age}" when older than one window	locale integer	same
Limitations	remaining observability flags as sentences (max 2 + "and N more")	bullets	same
Readiness banner	failing readiness checks: Blocking (registry, monitoring, jobs, PV/SP/OP/MODE tags), Configuration, Waiting for data; plant facts listed apart	"N to fix · N plant fact(s)"	`readiness`
PV / SP / OP / MODE tiles	live → newest trend bucket → last stored value	2 dp; source line "live (RBE)" / "last stored {age}"	live + historian

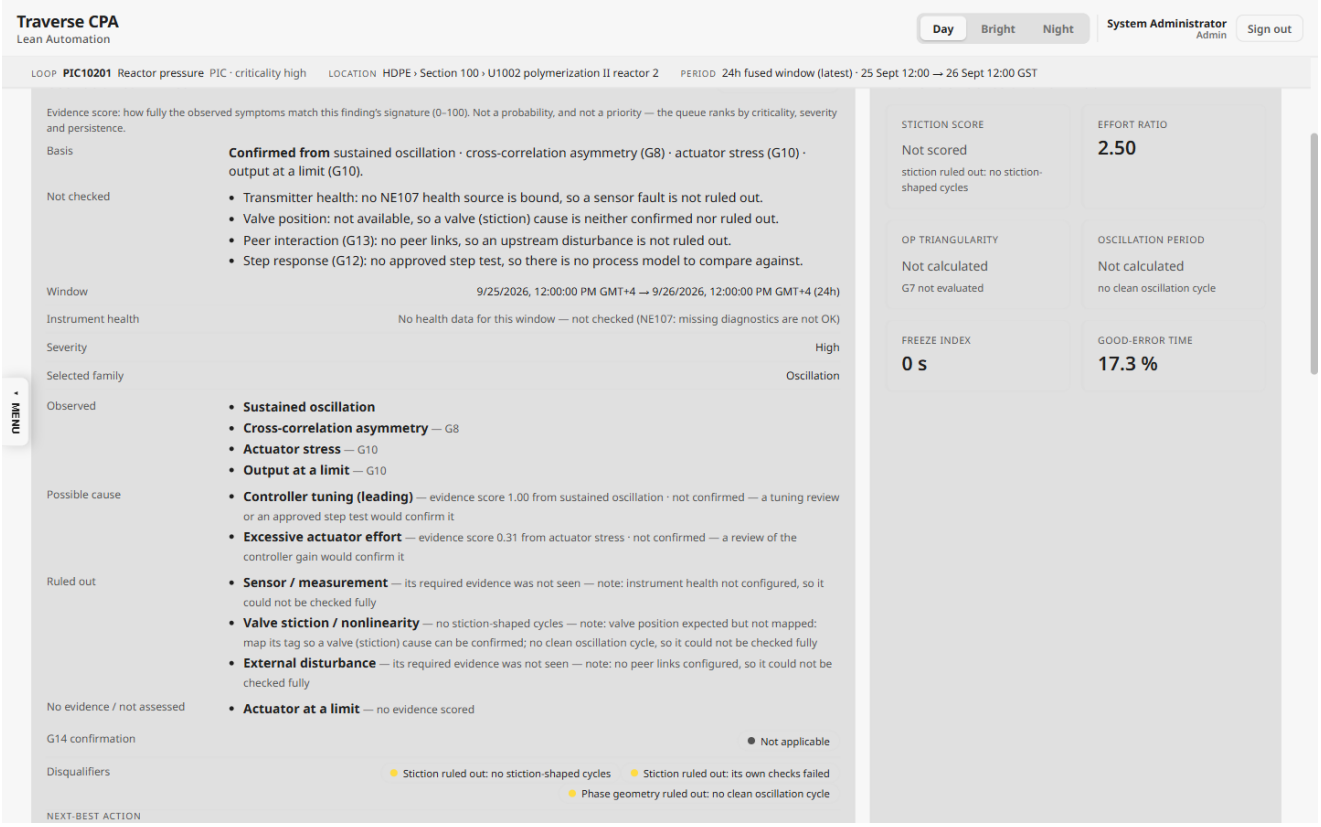


Loop Explorer — gate strip by tier with the mark legend

11.4 Investigation (/cpm/investigation)



Investigation — final conclusion

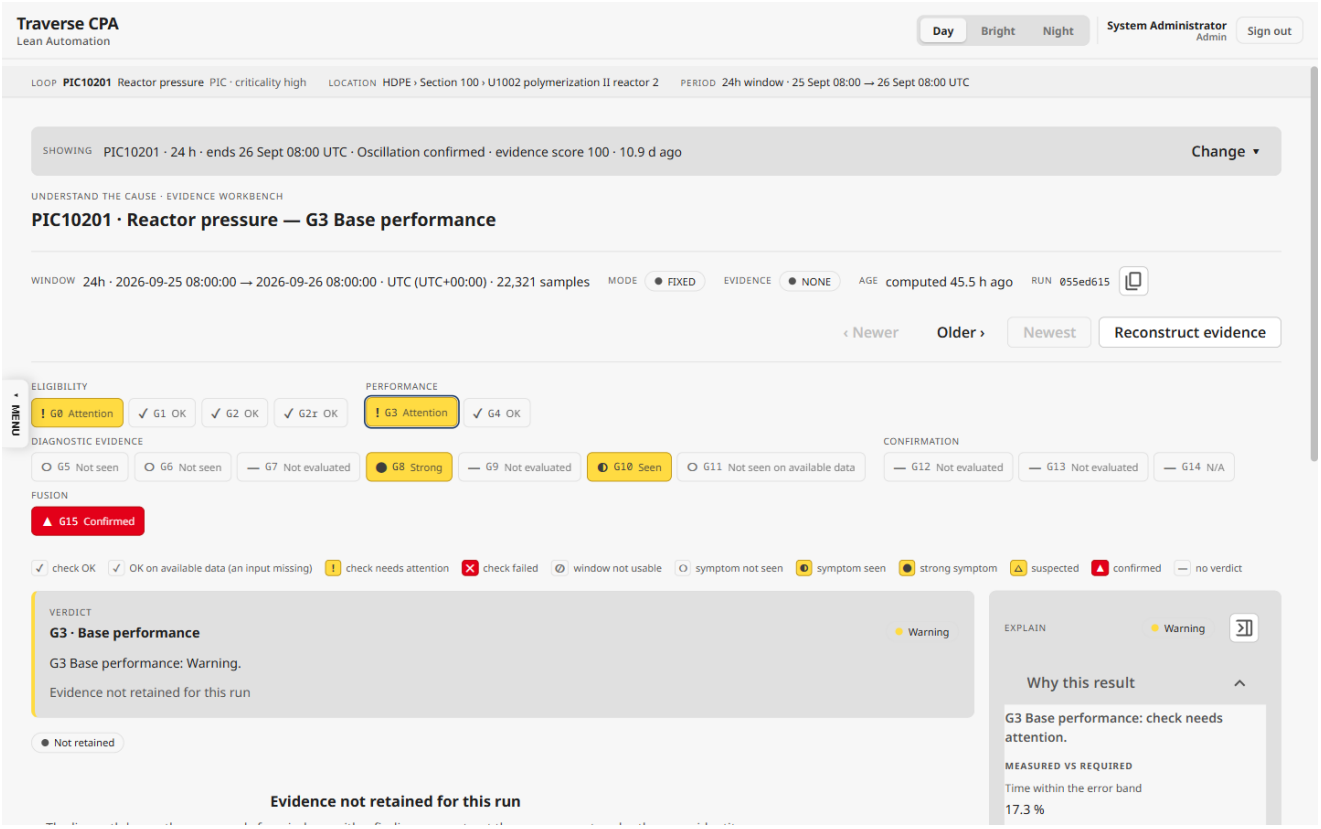


Investigation — basis, cause breakdown and key-fact tiles

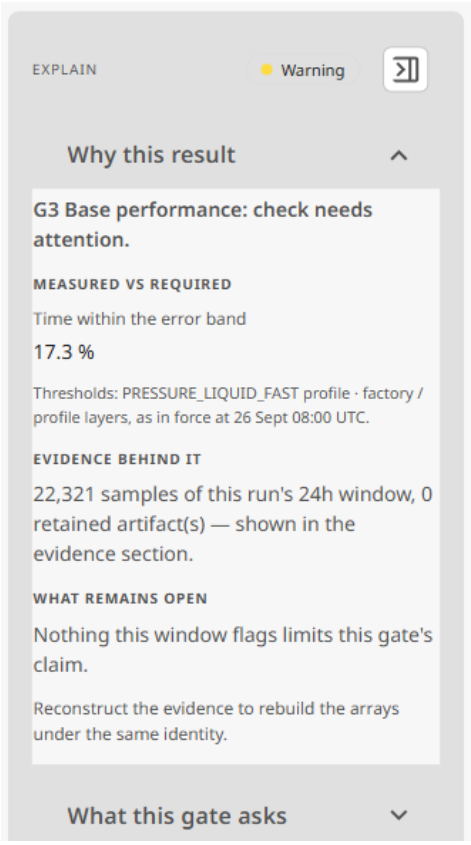
KPI on screen	Logic / formula	Display and thresholds	Endpoint
Evidence score pill + note	round(confidence × 100) when the diagnosis is a verdict, with the meaning and any cap	"evidence score N"	`gates/latest`
Observed / Possible cause / Ruled out / No evidence	v2 verdict: present observations; QUALIFIED hypotheses (score desc) with how to confirm; DISQUALIFIED with reasons; the rest	hypothesis score 0–1 (2 dp)	same
G14 confirmation	Confirmed / Contradicted / Not confirmed / Not applicable	Confirmed red, Contradicted green	same
Stiction score	stiction_score (= triangularity)	2 dp; "Not scored" when stiction is disqualified	same
Effort ratio	σ(OP)/σ(PV)	2 dp	same
OP triangularity	mean per-cycle SSE_sine/(SSE_sine + SSE_triangle)	2 dp; "Not calculated" without G7	same
Oscillation period	acf_period_s	s / min / h; "Not calculated — no clean oscillation cycle" when ≤ 0	same
Freeze index	freeze_index_s	duration	same
Good-error time	good_error_pct × 100	1 dp %; "N/A" on an excluded window	same
Window browser Δ	current − previous window: evidence score in points, key facts in their units	"+N pt"	`gates` history

Availability rule for every metric (`metricAvailability.ts`, first match wins): excluded window → N/A; insufficient → Insufficient data; G0 unusable → Insufficient data; missing → Not calculated; stiction disqualified → Not scored; period ≤ 0 → Not calculated; method not qualified → Not calculated / Not qualified; owning gate has no verdict → Not calculated; otherwise the measured number.

11.5 Evidence Workbench (/cpm/evidence)



Evidence Workbench — one method run, gate chips, verdict and Explain

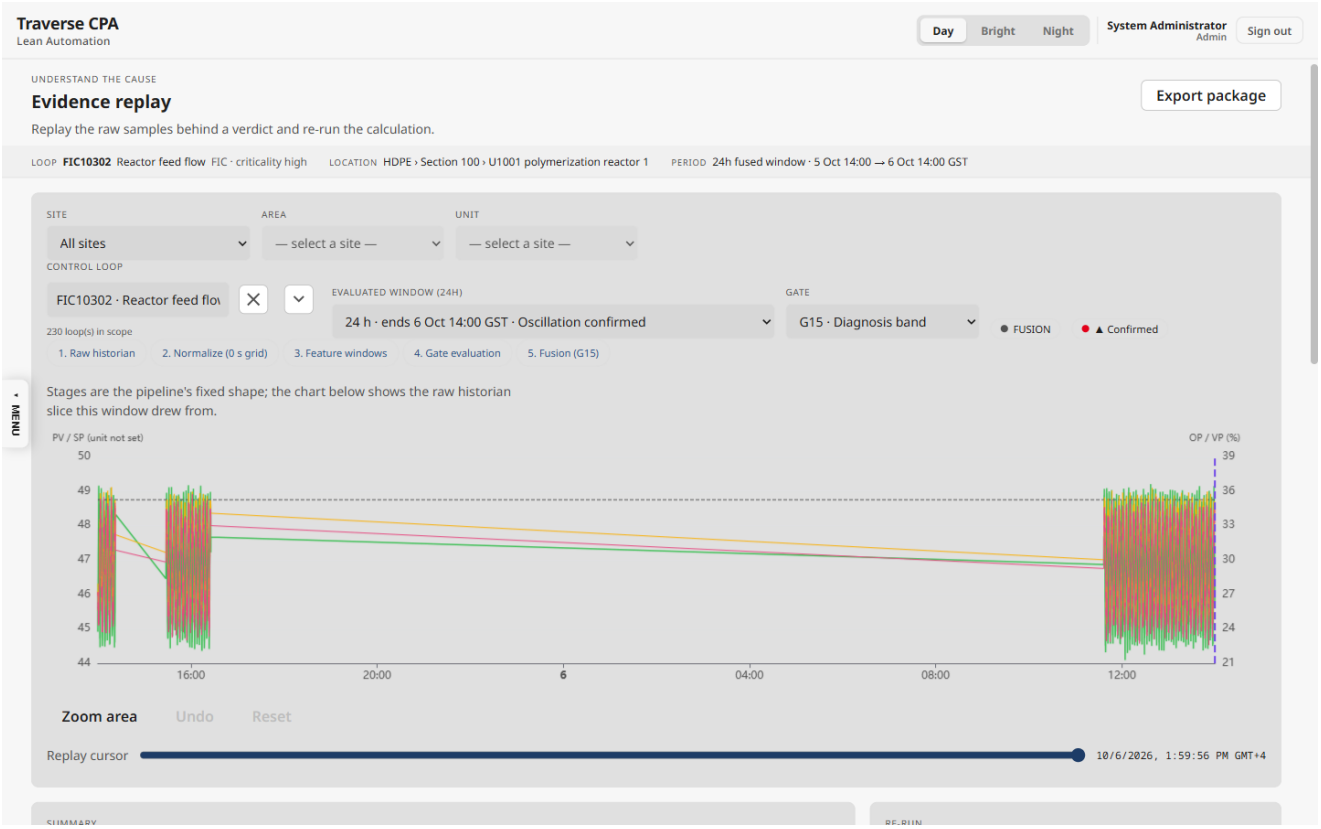


Explain › "Why this result": measured vs required, with the profile in force

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Window / samples / age	run support from → to, sample count, now − computedAt	site time zone	`method-runs/{id}`
Evidence readiness	NONE / COMPLETE / PARTIAL / PENDING / FAILED / EXPIRED (every chunk present and SHA-256 verified)	pill	`method-runs/{id}/evidence`
Headline tiles	per-gate provenance numbers, e.g. G0 accepted samples a of b; G5 ACF period, regularity, cycles; G9 cycle polygons + mean area	value + unit	evidence chunks
Why this result	measured value vs the threshold in force at the run's time (engine copy, else effective configuration)	✓ / ✕ per requirement	`configuration/effective?at=`
G15 why	evidence score of the selected family; agreeing windows vs required (persistence)	"N of 100"	fused row

11.5b Evidence Replay (/cpm/replay)

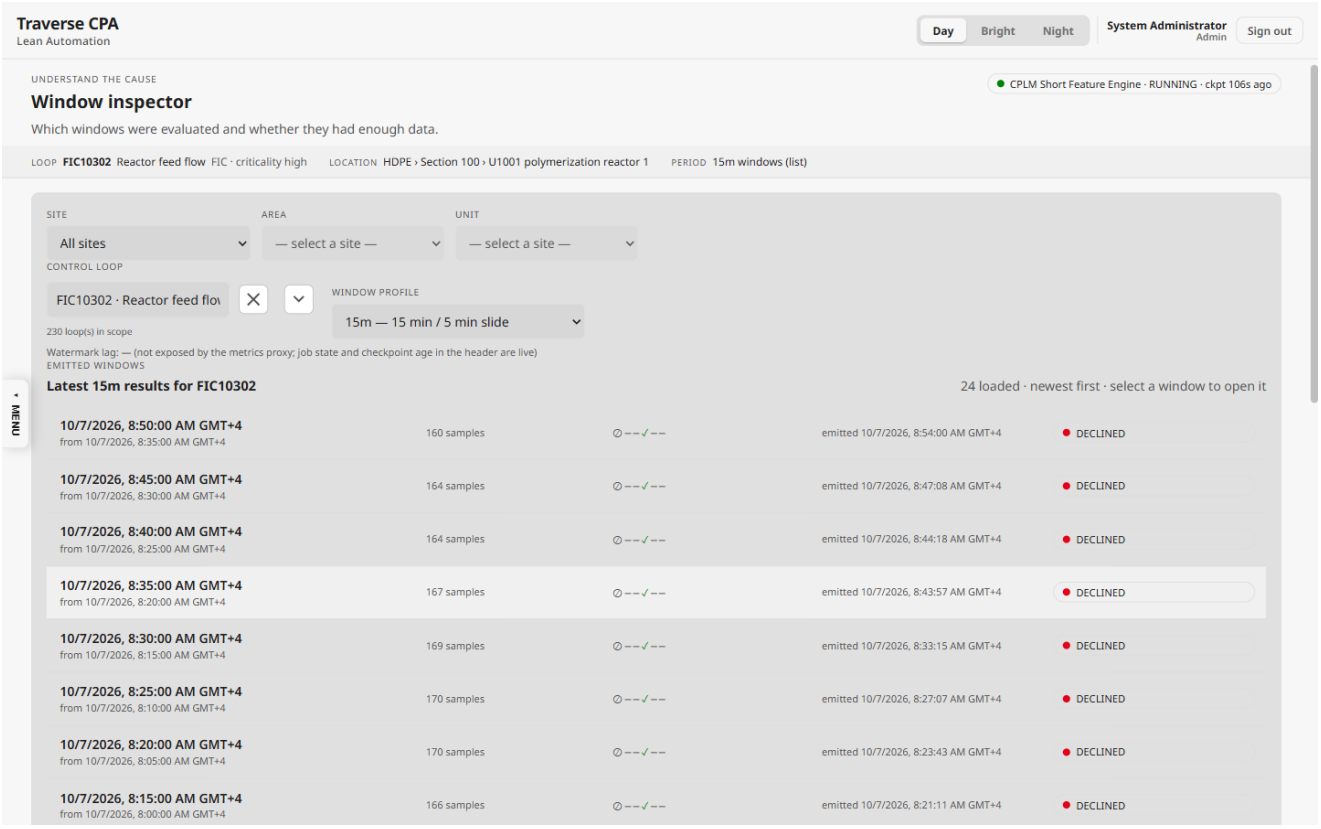
The raw historian samples behind one fused 24h verdict, and a re-run of the calculation.



Evidence Replay — raw PV / SP / OP / VP slice of the evaluated window with the replay cursor

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Evaluated window / gate	fused 24h windows (newest first) and the gate result for the selected gate	"24 h · ends …"; gate mark	`loops/{id}/gates`
Summary metrics	the long-tier (24h) row of the same window: completeness, AUTO fraction, MAE, IAE, good-error, effort, reversals, travel, ACF period / regularity, harmonics, triangularity, oddness, corner score	raw value, 3 dp (fractions not × 100)	`kpis?resolution=24h`
Raw slice	historian raw points for the window (PV, SP, OP, VP); shape gates G7–G10 also as a PV–OP phase plane	2 dp; truncated note at the point budget	`/api/hist/raw/cursor`
Re-run	recompute the loop through Flink and poll the replay until finished	FINISHED / FAILED	`loops/{id}/recompute`

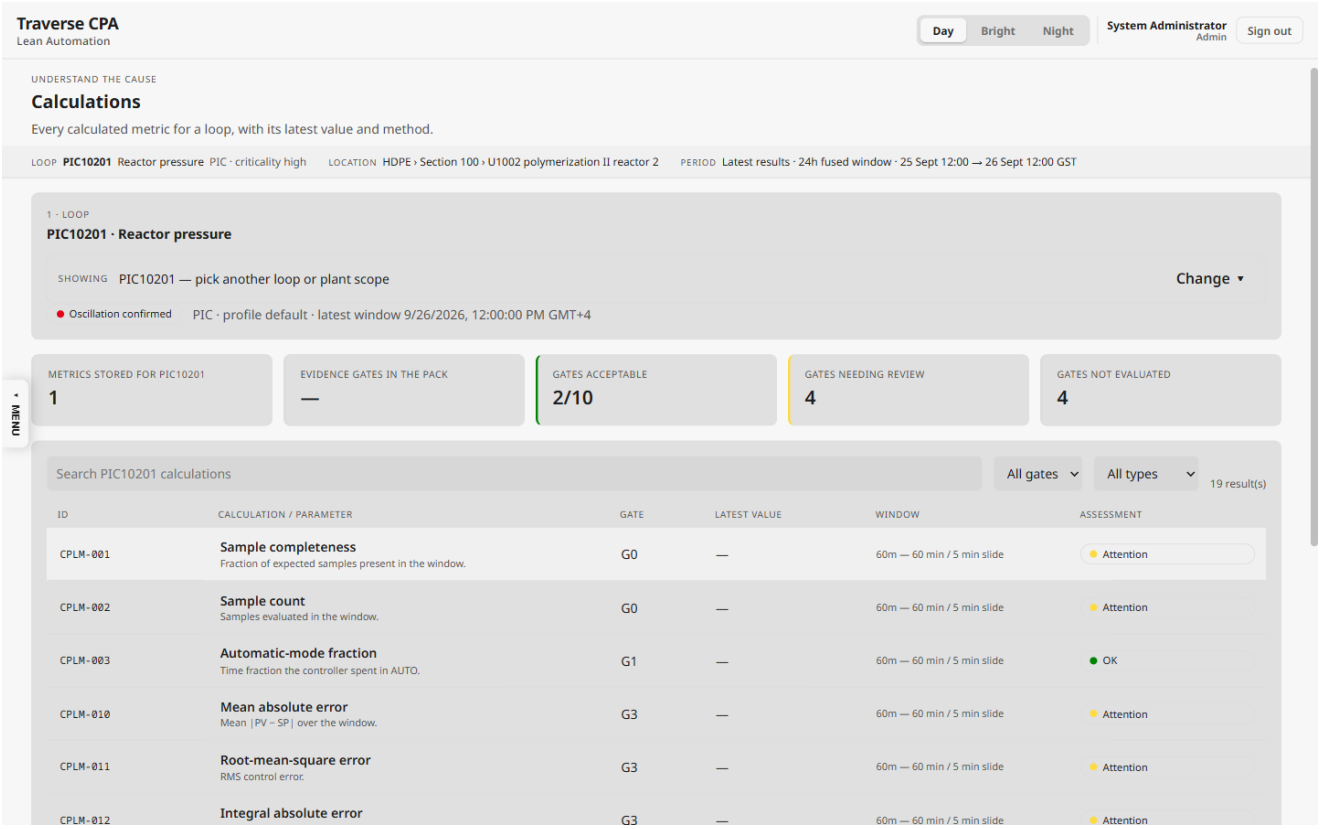
11.6 Window Inspector (/cpm/windows)



Window Inspector — emitted short windows with completeness and gate glyphs

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Completeness pill	completeness, else sample_count / expected (window / grid, 5 s fallback); DECLINED when sufficient_data = false	"N% full"; ≥ 90 green, ≥ 50 amber, < 50 red	`kpis?resolution=15m`
Short gate glyphs	G0, G1, G2, G2r, G3, G4 status per window	gate marks	same
Metrics	MAE, RMSE, IAE, ISE, good-error %, effort ratio, OP travel/day, reversals/h, in AUTO %, completeness %	fractions × 100 (1 dp %); else 4 significant digits	same
Sample density	24 buckets: raw PV points per bucket / expected per bucket	bar; < 0.6 sparse	historian raw
Compare across window sizes	per short kind, last 12 qualified windows: latest, median, range, count, declined	—	`kpis/latest`
Runtime chip	CPLM job state + last checkpoint age	RUNNING green	`pipeline-metrics`

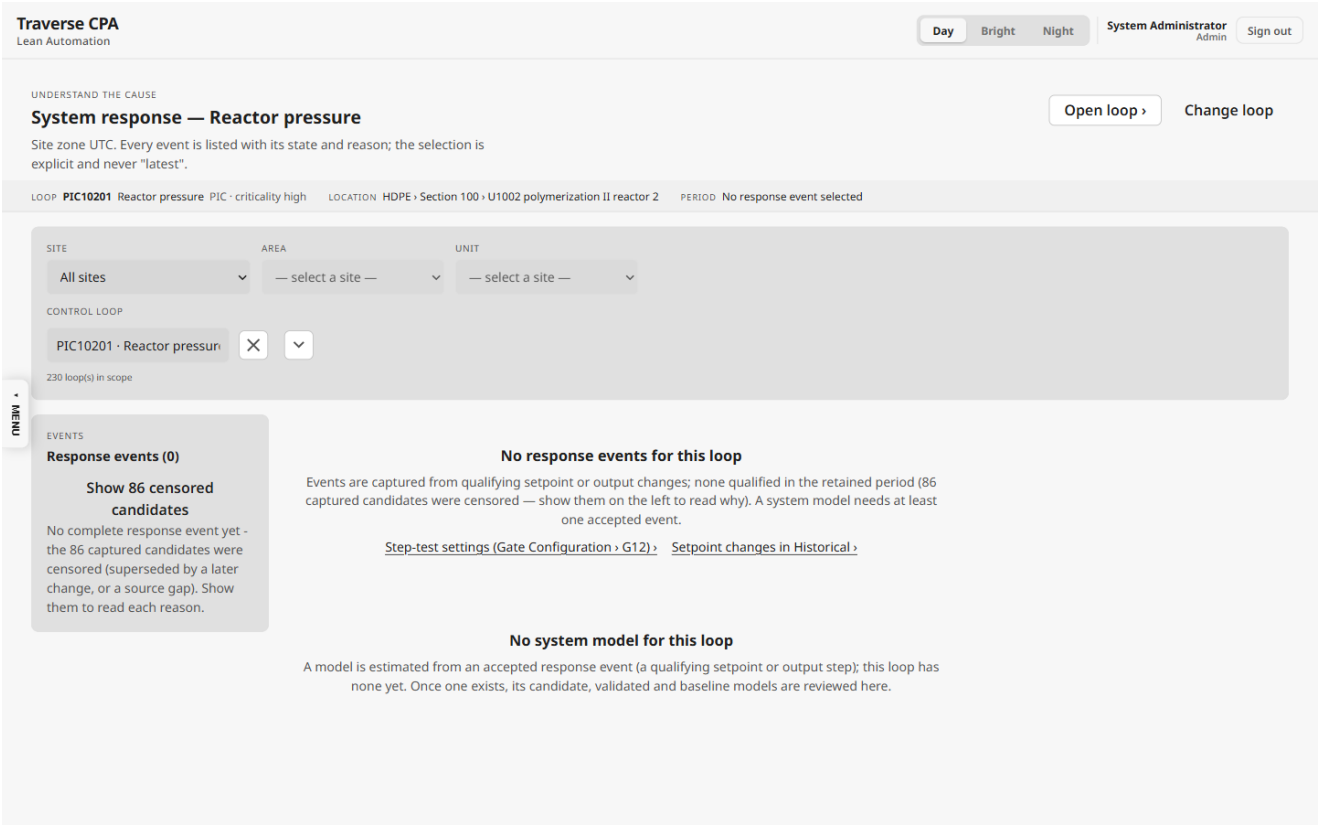
11.7 Calculations (/cpm/calculations)



Calculations — every calculated metric for a loop, its latest value, window and assessment

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Metrics stored	catalogue metrics with a value (latest 60m short row, latest 24h long row, fused confidence)	integer of 19	`kpis`, `gates/latest`
Gates acceptable g/t	distinct catalogue gates (t = 10) whose assessment is green	"g/t"	`gates/latest`
Gates needing review / not evaluated	assessment amber or red / grey	counts	same
Latest value	ratio / fraction 3 dp; count integer; seconds 1 dp; others 4 significant digits + unit; G15 shown as N / 100	availability word when not measured	same
Evidence-score band (G15 drawer)	first band with value ≤ max: NO_CALL 0.35, DETECTED 0.55, CLASSIFIED 0.75, SUSPECTED 0.90, CONFIRMED 1.00	band name	`calculations`

11.8 System Response (/cpm/response)



System Response — response events, censored candidates and the system model

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Response events (N)	captured G12 events, newest first, censored ones hidden unless shown	count	`loops/{id}/response-events`
Censored candidates	captures with capture_state = CENSORED (superseded, gap, mode change, too long)	count + reason	same
Model card	family (FOPDT / IPDT) · model state; Kp, θ, τ (± std. error)	2 dp or exponential	`response-events/{id}`
Key results	t63, rise time 10–90, settling, overshoot, steady error, IAE / ISE / ITAE	value ± uncertainty + unit	same
Validation (subgate E)	fit improvement, free-run RMSE, bias, whiteness	3 dp	same
Model history	every model per parameter; review sets VALIDATED / REJECTED / ACTIVE_BASELINE	chart + table	`loops/{id}/models`

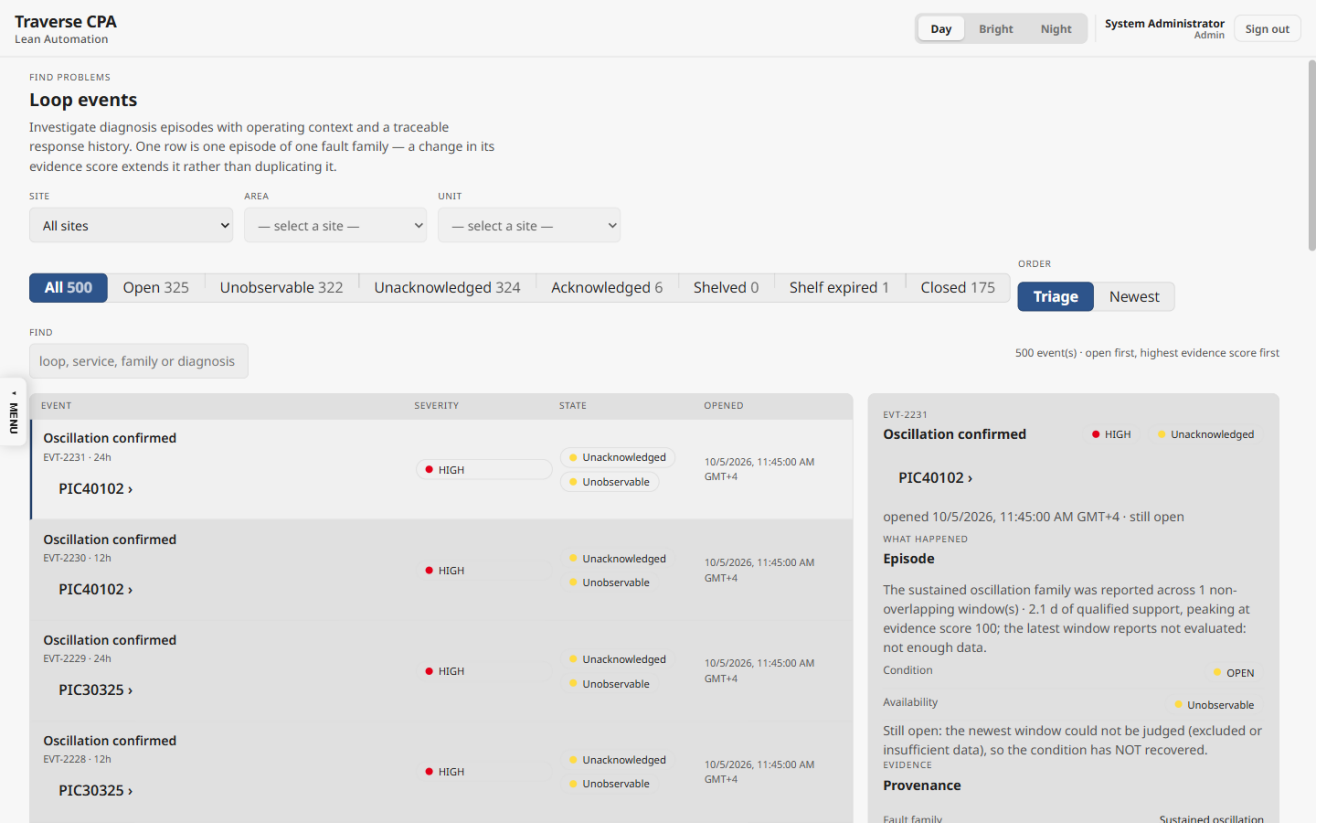
11.9 Historical (/cpm/historical)



Historical — signals, KPI overlay, mode and verdict on one time axis

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Signal lanes	historian PV (min–max envelope), SP, OP, VP; optional normalise ÷ own span	2 dp; PV unit from registry	`/api/hist/trend`
KPI overlay	one of effort ratio (15m), IAE (15m), good-error time (15m), triangularity (24h), harmonic energy ratio (24h) over the range	raw value (good-error as 0–1)	`kpis` (≤ 500 rows)
Mode lane	automatic / manual segments from the mode signal	green / amber	historian
Verdict lane	24h fused verdicts: Suspected (CONFIRMED / SUSPECTED), Developing (DETECTED / CLASSIFIED), Normal, Not evaluated	red / amber / green / grey	`gates` history (≤ 100)
Selected window	verdict, evidence score, samples, from / to, calculation and profile versions	—	same

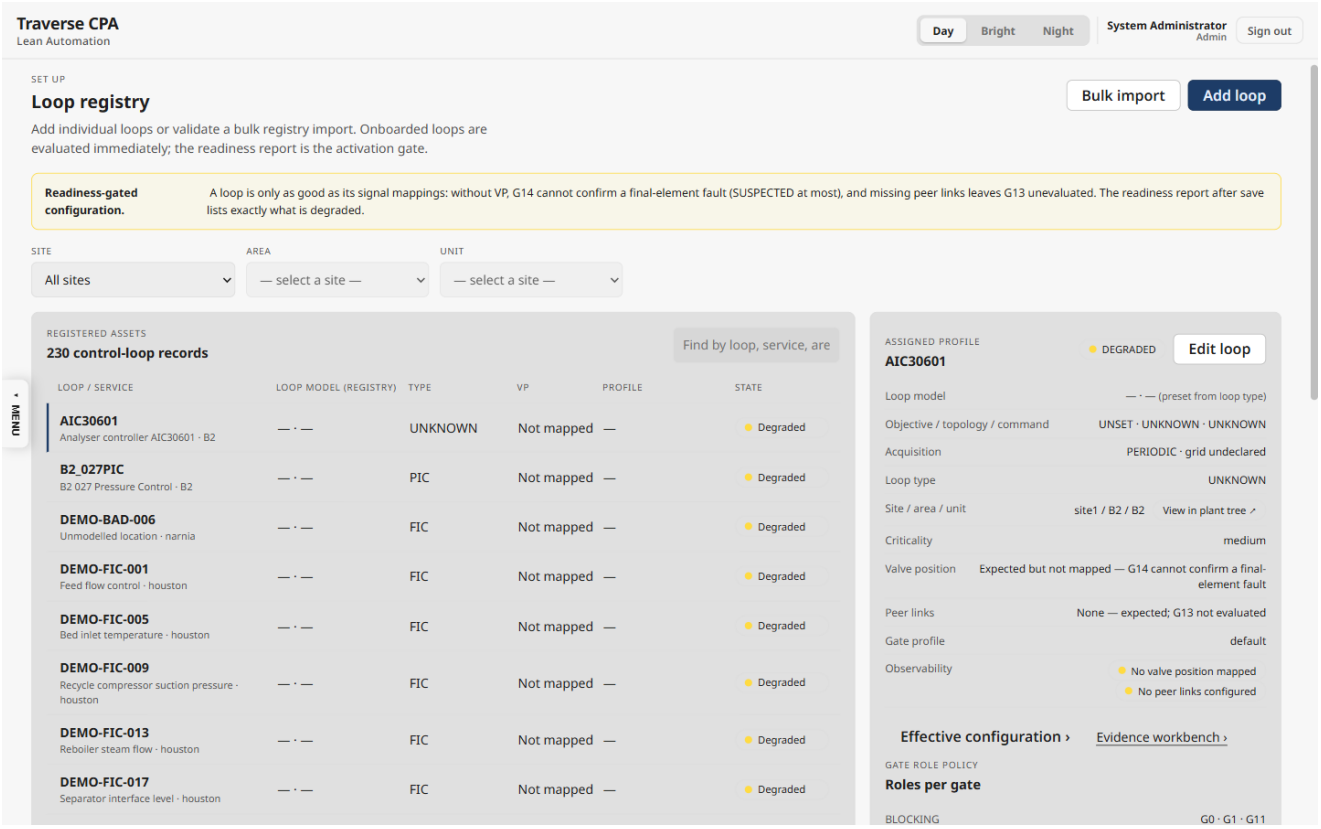
11.10 Loop Events (/cpm/events)



Loop Events — server-paged episodes with tab counts

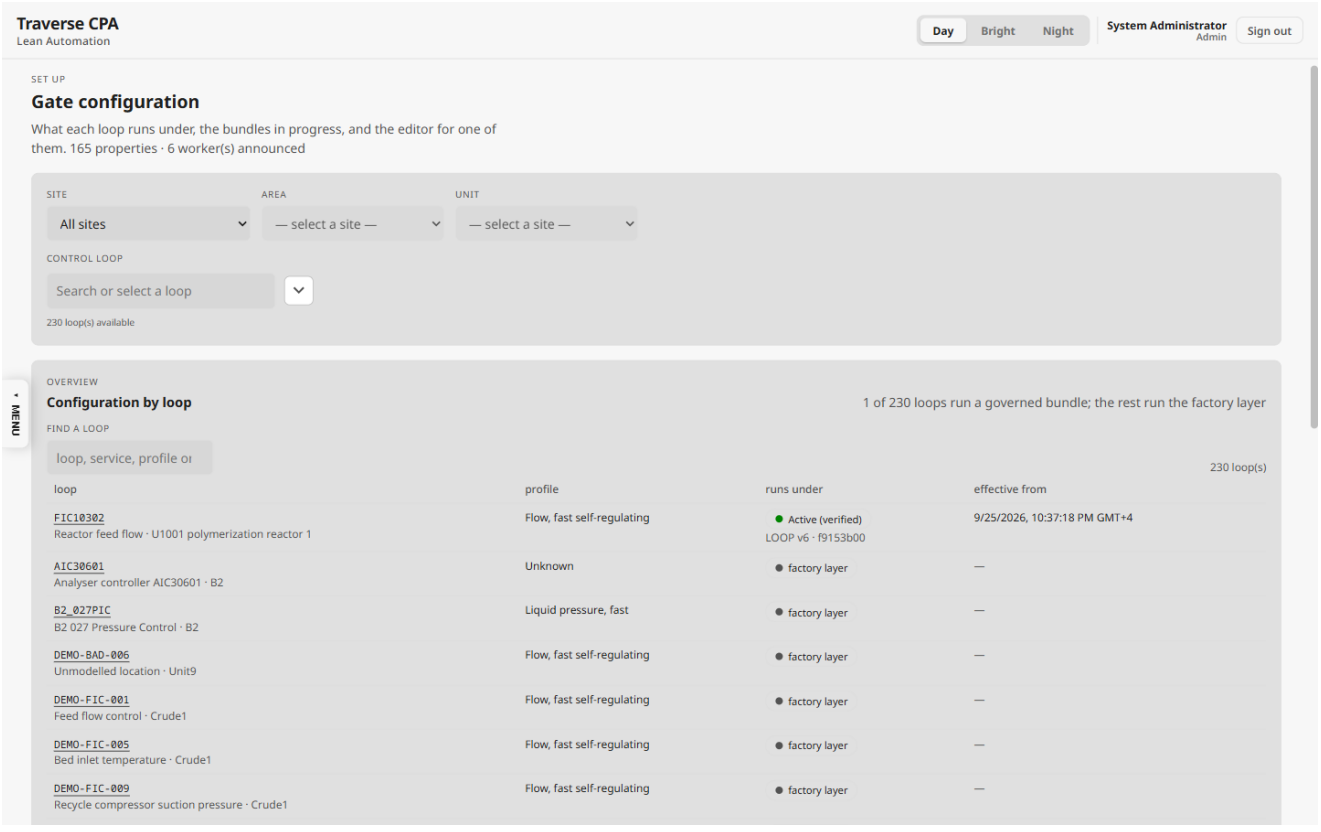
KPI on screen	Logic / formula	Display and thresholds	Endpoint
Tab counts	All; Open (not closed); Unobservable (open + UNOBSERVABLE); Unacknowledged (open, not acked, not actively shelved); Acknowledged; Shelved (shelve_until > now); Shelf expired; Closed	counts under the same loop / scope / search	`events/page`
Order	Triage: open first, highest peak evidence score, newest; or newest first	caption	same
Severity	episode severity band (latest of its own family); fallback from peak score ≥ 0.9 / ≥ 0.55	HIGH red, MEDIUM amber, LOW green	same
State	Closed · Closed unacked · Shelved (expires in …) · Shelf expired (unreviewed) · Acknowledged · Unacknowledged	pill	same
Episode sentence	window_count non-overlapping windows · qualified support (h / d) · peak evidence score	1 dp	same
Conditions (G10 / G11)	fast limit / freeze conditions; stale when open with no update for 10 min	duration, since	`conditions`

11.11 Loop Registry, Gate Configuration, Pipeline Health



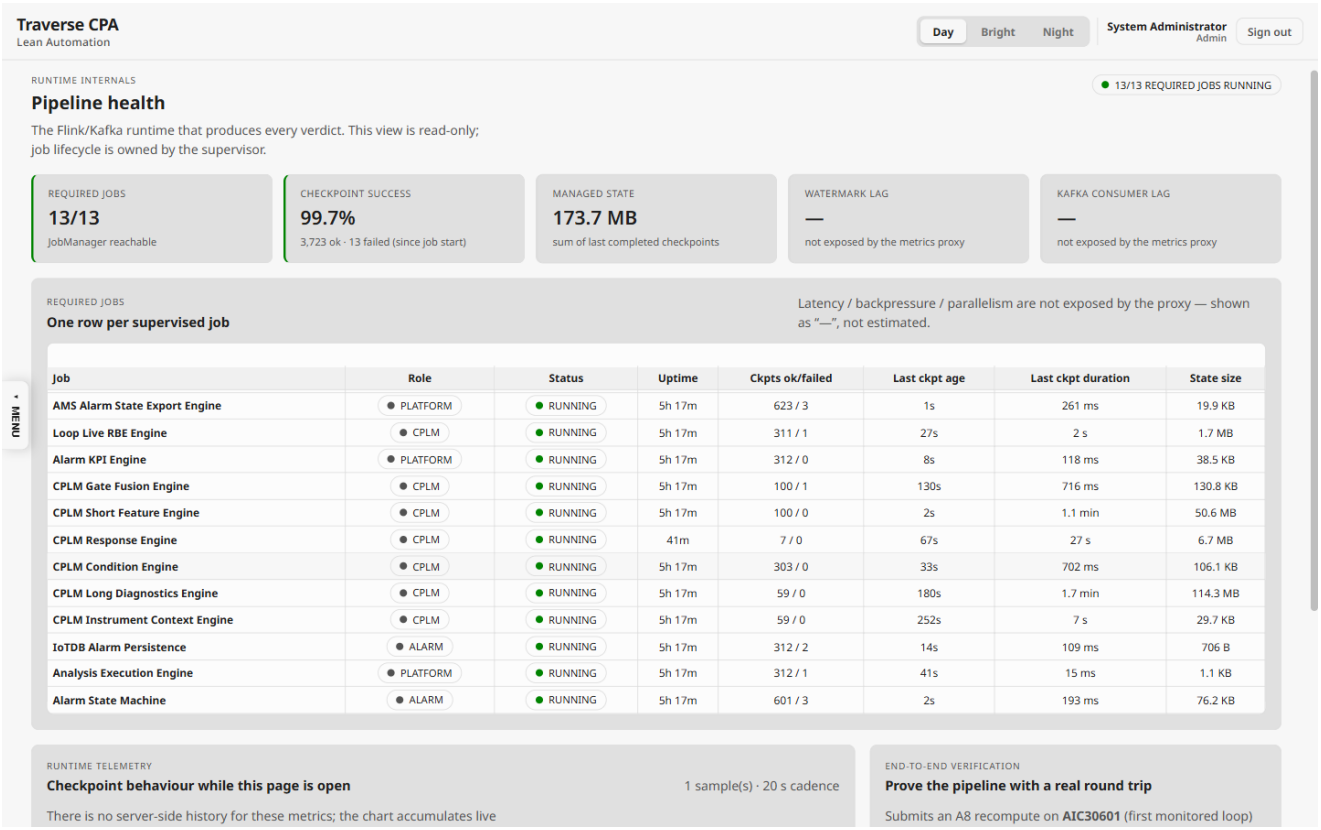
Loop Registry — loop records, VP state and assigned profile

KPI on screen	Logic / formula	Display and thresholds	Endpoint
State	Inactive / Registered (not monitored) / Degraded (NO_VP or no links) / Active	pill	`loops`
VP column	Mapped / Not mapped / Not installed / No valve (applicability)	word	same



Gate Configuration — which configuration layer each loop runs under

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Governed loops	loops whose LOOP / PROFILE / SITE bundle is active (first hit wins), else factory layer	"N of M loops run a governed bundle"	`configuration/effective/summary`
Layer stack	properties per source layer (factory / profile / site / loop)	counts	`configuration/effective`
Shadow comparison	symptoms removed / added per gate between a draft and LIVE results	counts	`drafts/{id}/shadow`



Pipeline Health — required jobs, checkpoints and the delivery path

KPI on screen	Logic / formula	Display and thresholds	Endpoint
Required jobs r/n	required Flink jobs RUNNING (n = 13)	green when all running	`pipeline-status`
Checkpoint success	100 × Σ completed / (Σ completed + Σ failed)	1 dp %; ≥ 99 green	`pipeline-metrics`
Managed state	Σ last checkpoint state size	B / KB / MB	same
Last fused verdict	now − newest fused window end	min / h	`fleet/rankings`

12. Calculation catalogue (CPLM IDs)

The identifiers used on the Calculations page (`calcCatalogue.ts`). "Source" says which tier stores the value: short = 1m–60m windows, long = 4h–24h windows, gate = fused verdict.

ID	Calculation	Field	Gate	Unit	Source	Definition
CPLM-001	Sample completeness	`completeness`	G0	ratio	short	Fraction of expected samples present in the window.
CPLM-002	Sample count	`sample_count`	G0	count	short	Samples evaluated in the window.
CPLM-003	Automatic-mode fraction	`auto_pct`	G1	ratio	short	Time fraction the controller spent in AUTO.
CPLM-010	Mean absolute error	`mae`	G3	PV units	short	Mean |PV − SP| over the window.
CPLM-011	Root-mean-square error	`rmse`	G3	PV units	short	RMS control error.
CPLM-012	Integral absolute error	`iae`	G3	PV·s	short	Accumulated absolute error.
CPLM-013	Integral squared error	`ise`	G3	$PV^2 \cdot s$	short	Accumulated squared error.
CPLM-014	Good-error time	`good_error_pct`	G3	fraction	short	Time fraction (0-1) the error stayed inside the good band.
CPLM-020	Actuator effort ratio	`effort_ratio`	G4	ratio	short	OP travel relative to the error it corrects.
CPLM-021	OP travel per day	`travel_per_day`	G4	%/day	long	Total actuator travel extrapolated to a day.
CPLM-022	OP reversals per hour	`reversals_per_hour`	G4	per h	long	Direction changes of the actuator.
CPLM-030	Oscillation period (ACF)	`acf_period_s`	G5	s	long	Dominant period from the autocorrelation.
CPLM-031	Oscillation regularity	`acf_regularity`	G5	ratio	long	How regular the oscillation is (0–1).
CPLM-040	Harmonic amplitude ratio	`harmonic_amplitude_ratio`	G6	ratio	long	Harmonics vs fundamental amplitude.
CPLM-041	Harmonic energy ratio	`harmonic_energy_ratio`	G6	ratio	long	Spectral energy in harmonics.
CPLM-050	OP triangularity	`triangularity`	G7	score	long	Triangular-wave similarity of the actuator trace (stiction shape).

ID	Calculation	Field	Gate	Unit	Source	Definition
CPLM-060	Horch oddness	`horch_oddness`	G8	score	long	Odd-symmetry of the PV–OP cross-correlation.
CPLM-070	Phase-portrait corner score (raw)	`corner_score`	G9	score	long	Sharp-corner evidence in the PV–OP phase plot. CAUTION: the raw statistic reads high (~0.9) even on healthy noisy data; treat it only alongside the G9 verdict and phase-area band, never alone.
CPLM-090	Fused evidence score	`confidence`	G15	score	gate	How fully the observed symptoms match the selected family (0–1; shown 0–100). Uncalibrated — not a probability.

13. Alarm KPIs (AMS, outside the CPA pages)

Computed by `AlarmKpiStreamJob` from alarm lifecycle events and pushed to the alarm dashboard over SignalR.

KPI	Formula	Window	Thresholds
Alarm rate	count of lifecycle events with state ACTIVE	10 min sliding, every 1 min	> 10 minor flood, > 20 major flood, > 50 severe flood (fixed)
Standing alarms	+1 on ACTIVE, −1 on CLEARED (floored at 0)	running count	—

14. Known gaps and inconsistencies

Found while documenting. None changes a stored value; they affect wording, consistency or configurability. Listed so readers do not mistake them for intended behaviour.

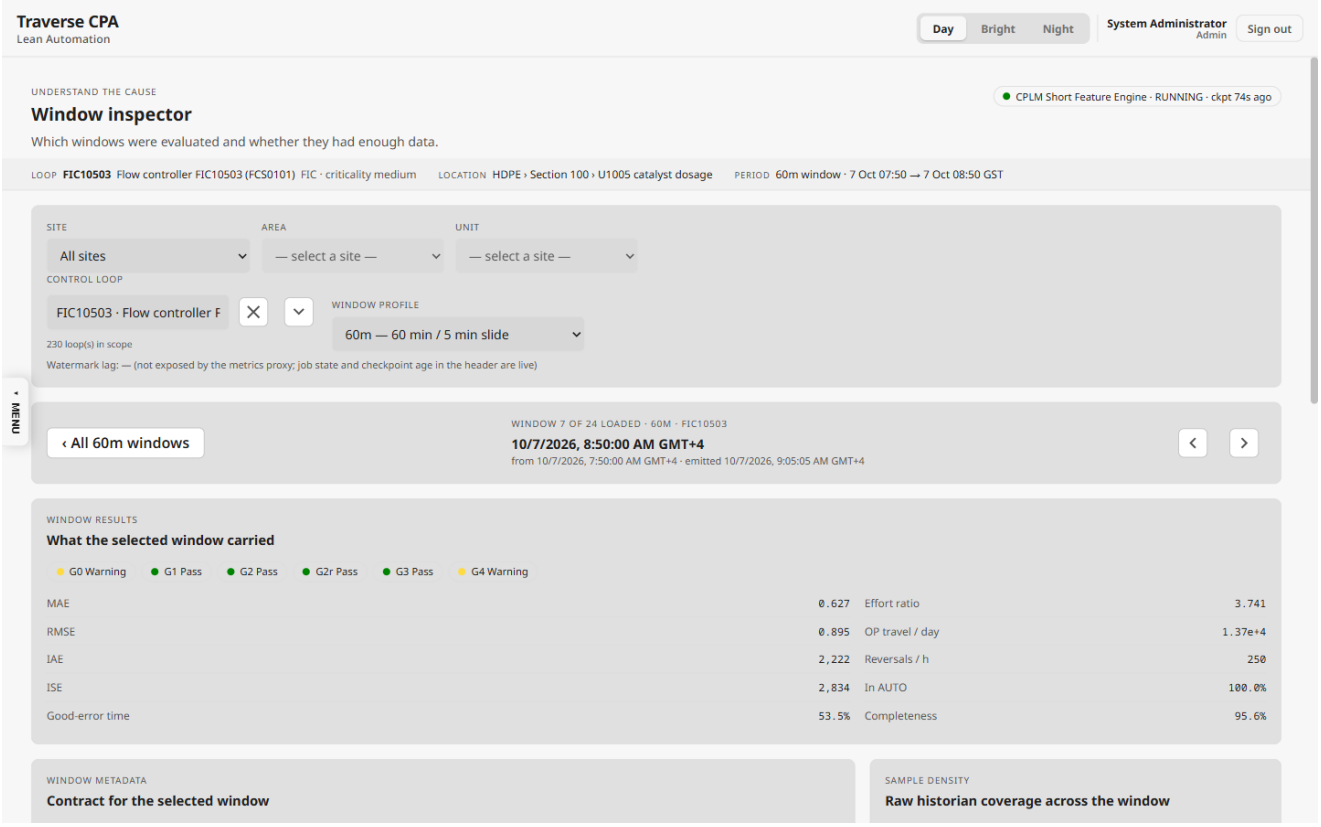
#	Area	Observation	Effect
1	Persistence	"2 of 3 windows" counts consecutive 15-min re-evaluations of a sliding 12h / 24h window — about 45 min of mostly identical data; the configuration key assumes 12 h per window	Persistence is weaker than its name
2	G5 status	WARN whenever any ACF period exists, even out of band or irregular (band and regularity only gate period validation)	G5 "Seen" more often than expected
3	G12 on the verdict	PASS only from the registry flag "has step test"; not joined to the Response Engine's results	G12 chip does not reflect captured step responses
4	Response Engine	Stability margins and SIMC advice never produced (no controller parameters); LIC set-point steps unsupported; trigger span fixed at 100 EU	Some System Response panels stay empty
5	Hard-coded thresholds	G0 0.98 / 0.50, G1 0.90, G2r 0.05 / 0.20, G6 0.5, G9 0.30, G11 60 s / 10 %, G13 xcorr 0.5, bands 0.35 / 0.55 / 0.75 / 0.90, caps 0.89 / 0.54	Not adjustable on Gate Configuration
6	Causes	ACTUATOR_LIMIT and SENSOR hypotheses are scored but never become the leading cause or the diagnosis	Shown only as possible causes
7	Display units	Hypothesis scores 0–1 (2 dp) vs evidence score 0–100; Explorer History "peak N%"; Replay and Historical overlay show good-error as 0–1	Same quantity in two scales
8	Historical verdict lane	CONFIRMED_* labelled "Suspected"	Under-states confirmed findings
9	Performance tiles	Health filter changes the tiles only; tiles count INSUFFICIENT and EXCLUDED loops as evaluated; "in scope" counts all registered loops in one place, monitored loops in another	Tile denominators differ from the matrix
10	Missing confidence	A missing fused `confidence` is stored as 0.0	Can read "evidence score 0"
11	Alarm KPIs	Standing count is a global counter not reconciled with state; oldest-standing duration always 0	Approximate standing count after restarts
12	Engine applicability	The fusion job does not read VP / peer applicability (CHG-059); the UI words the flag per loop	A valve cause can still rank on a declared "no valve" loop
13	Replay stepper	"Normalize (0 s grid)" when the window row carries no sample period	Cosmetic

Appendix A — Source index

Topic	Source files (repo-relative)
Short KPIs, G0–G4	`src/flink/.../cplm/engine/ShortFeatureCalculator.java`, `SignalMath.java`, `LimitMath.java`, `ShortWindowSpecs.java`
Long diagnostics G5–G11, G13	`.../cplm/engine/LongDiagnosticsCalculator.java`, `OscillationMath.java`, `WelchPsd.java`, `ShapeMath.java`, `VpTrackingMath.java`, `InteractionGate.java`, `InteractionMath.java`
Fusion G14 / G15	`.../cplm/CplmGateFusionEngine.java`, `fusion/FusionEvidence.java`, `fusion/Confirmation.java`, `fusion/DiagnosisBands.java`, `fusion/VerdictAssembly.java`, `fusion/GeometryEvidence.java`
Defaults	`.../cplm/CplmLoopDynamicsProfilePacks.java`, `resources/cplm/loop-dynamics-profiles.yaml`, `resources/cplm/gate-config/factory-layer.json`, `config/BundleProfileMapper.java`
G12 response	`.../cplm/response/*` (ResponseEventDetector, ResponseCapture, metrics/*, estimation/*, validation/ModelValidator)
Instrument health	`.../cplm/health/HealthEligibilityPolicy.java`, `DeviceStateMachine.java`
Episodes	`src/services/cplm-api/Services/EventFrameLifecycle.cs`, `BackgroundServices/CplmEventFrameService.cs`, `CplmShelfExpirySweeper.cs`
Fleet KPIs	`src/services/cplm-api/Data/FleetLatestSql.cs`, `FleetPrioritySql.cs`, `EventPageSql.cs`, `Controllers/CpmFleetController.cs`, `Services/Registry/OptionalInputs.cs`
Display rules	`src/frontend-ob/src/components/Cpm/gateStatus.ts`, `evidenceScore.ts`, `metricAvailability.ts`, `conclusionBasis.ts`, `causeAnalysis.ts`, `calcCatalogue.ts`, `windowLabel.ts`
Alarm KPIs	`src/flink/.../AlarmKpiStreamJob.java`

Appendix B — Short-window field dictionary (1m … 60m)

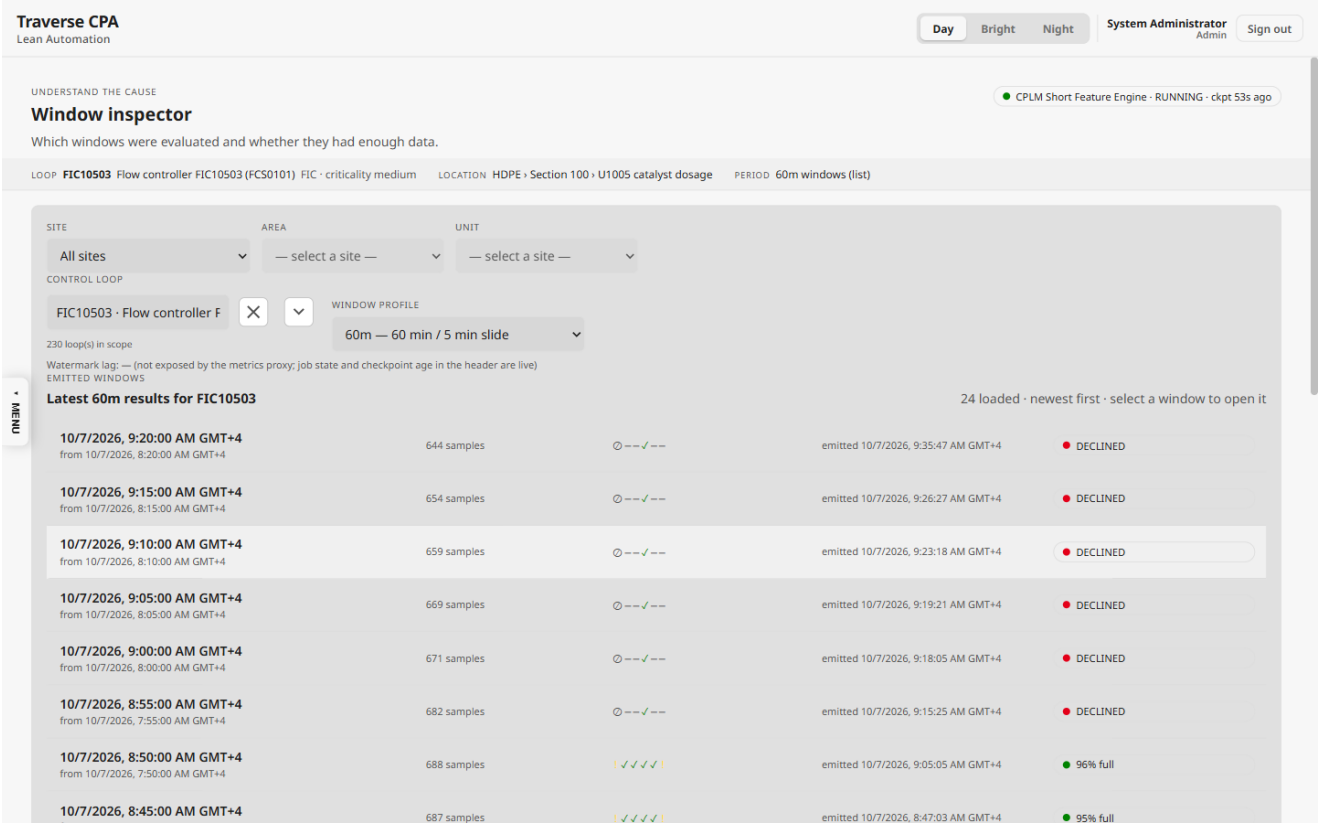
Every value stored for each short window (`cplm_short_feature_results`, topic `feature.short.v1`). On screen: WI = Window Inspector · CALC = Calculations page · EXP = Explorer › Show calculations · INV = Investigation key facts · HIST = Historical KPI overlay · REP = Evidence Replay summary · WB = Evidence Workbench · API = stored and served by `/loops/{id}/kpis`, not shown as its own number.



Window Inspector — one 60m window of FIC10503: G0–G4 statuses and the short-window KPIs

Field	Gate	Formula / definition	Unit / range	On screen
`expected_sample_count`	G0	max(1, round(window s / grid))	count	WI
`sample_count`	G0	samples with t in [start, end)	count	WI, CALC (CPLM-002), REP
`completeness`	G0	sample_count / expected_sample_count (can exceed 1)	fraction	WI, CALC (CPLM-001), REP
`bad_quality_pct`	G0	count(quality ≠ GOOD) / n	0–1	API
`unknown_quality_pct`	G0	count(quality = UNKNOWN) / n	0–1	API
`duplicate_timestamps`	G0	count of repeated timestamps (must be 0 for PASS)	count	API
`sampling_jitter`	G0	σ(Δt) / median(Δt)	ratio	API
`gap_count`, `max_gap_s`	G0	intervals Δt > 1.5·Ts; the longest Δt	count, s	API

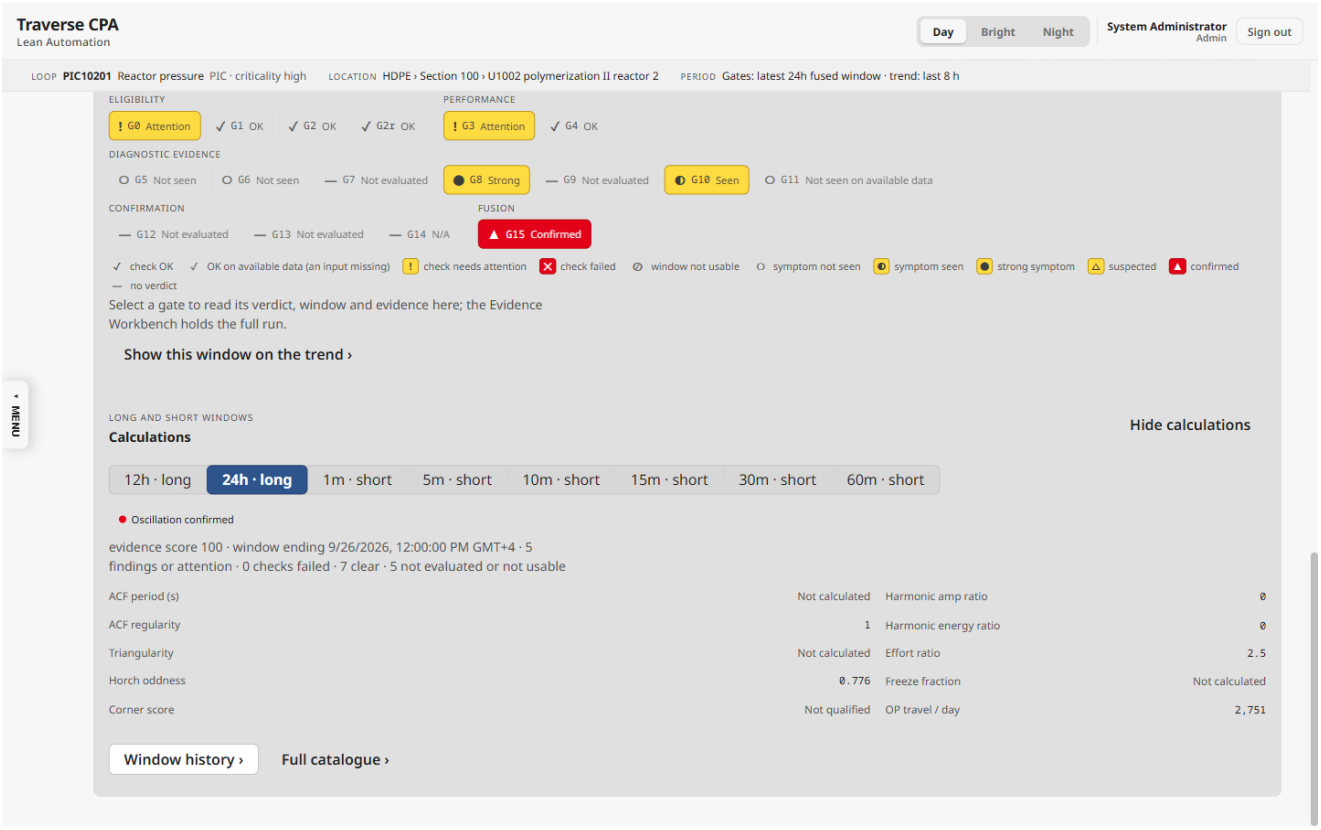
Field	Gate	Formula / definition	Unit / range	On screen
gate0_status / reason	G0	FAIL bad ≥ 0.50 · PASS completeness ≥ 0.98 ∧ bad < 0.05 ∧ no duplicates · WARN completeness ≥ 0.95 · else FAIL; reason BAD_QUALITY / QUALITY_UNKNOWN / INCOMPLETE	enum	WI glyph, EXP, WB
auto_pct	G1	count(mode automatic) / n	0–1	WI (In AUTO %), CALC (CPLM-003), REP
manual_pct	G1	1 − auto_pct	0–1	API
mode_changes_per_h	G1	mode changes / window hours	1/h	API
gate1_status	G1	PASS ≥ 0.90 · WARN ≥ 0.70 · else EXCLUDED	enum	WI glyph, EXP, WB
sp_min, sp_max, sp_range	G2	min, max and max − min of SP	PV units	API
sp_changes_per_h	G2	count(|ΔSP| > 1e-6) / hours	1/h	API
gate2_band_eu, band source	G2	declared band, else span fraction × PV span (0.01 tight / 0.05 averaging), else 1.0	PV units	API
gate2_status	G2	PASS if sp_range < band, else WARN	enum	WI glyph, EXP
region_out_of_band_pct	G2r	share of samples outside PV / OP region bounds	0–1	API
gate2r_status, operating_region_valid	G2r	PASS ≤ 0.05 · WARN ≤ 0.20 · else FAIL (invalid)	enum, bool	WI glyph, EXP
mae	G3	mean |SP − PV|	PV units	WI, CALC (CPLM-010), REP, compare table
rmse	G3	sqrt(mean (SP − PV)²)	PV units	WI, CALC (CPLM-011)
iae	G3	Σ |e|·Δt	PV·s	WI, CALC (CPLM-012), HIST, REP
ise	G3	Σ e^2·Δt	PV^2·s	WI, CALC (CPLM-013)
itae	G3	Σ (t − start)·|e|·Δt	PV·s^2	API
good_error_pct	G3	count(|e| ≤ 0.005 × PV span) / n (0.5 EU band without a declared range)	0–1 (shown ×100 %)	WI, CALC (CPLM-014), HIST, INV, Overview / Performance
oce	G3	auto_pct × good_error_pct × (1 − saturation_pct)	0–1	API
gate3_status	G3	PASS if good_error_pct ≥ 0.50 (0.30 averaging), else WARN	enum	WI glyph, EXP, WB
pv_std, op_std	G4	σ(PV), σ(OP)	PV units, %	API
effort_ratio	G4	σ(OP) / σ(PV)	ratio	WI, CALC (CPLM-020), HIST, INV
effort_ratio_normalized	G4	(σOP / OP observed range) / (σPV / PV observed range)	ratio	API
activity_ratio_norm, basis	G4	(σOP/100) / (σPV / declared PV span); null without a declared range (then G4 judges effort_ratio)	ratio	WB (Why G4)
op_travel	G4	Σ |ΔOP|	%	API
travel_per_day	G4	op_travel × 24 / window hours	%/day	WI, CALC (CPLM-021), REP
reversal_count, reversals_per_hour	G4	OP direction changes; ÷ hours	count, 1/h	WI, CALC (CPLM-022), REP
saturation_pct	G4	count(OP ≤ 5 % or ≥ 95 %) / n	0–1	API
freeze_index_s (short)	G4	longest run of unchanged PV × Ts	s	API
gate4_status	G4	STRONG activity > 8 · WARN > 3 · else PASS	enum	WI glyph, EXP, WB
sufficient_data	—	false when G0 FAIL or n < 10 (metrics null)	bool	WI (DECLINED)



Window Inspector — the emitted 60m windows: samples, G0–G4 glyphs and completeness (DECLINED when not qualified)

Appendix C — Long-window field dictionary (4h / 12h / 24h)

Every value stored for each long window (`cplm_long_feature_results`, topic `feature.long.v1`); the long record also embeds the aligned short features (Appendix B) for the same window. On screen: WI = Window Inspector · CALC = Calculations page · EXP = Explorer › Show calculations · INV = Investigation key facts · HIST = Historical KPI overlay · REP = Evidence Replay summary · WB = Evidence Workbench · API = stored and served by `/loops/{id}/kpis`, not shown as its own number.



Loop Explorer › Show calculations — the 24h long-window metrics behind a fused verdict, with availability words (Not calculated / Not qualified)

Field	Gate	Formula / definition	Unit / range	On screen
`method_eligibility.{G5,G6,G12}`	G5/G6/G12	grid ≤ τmin / 8 for G5, G6 (≤ τmin / 4 for G12): ELIGIBLE, else LIMITED (bandwidth)	enum	WB, flag BANDWIDTH_LIMITED
`acf_gamma0`	G5	Σ x² / n (detrended shape signal)	signal²	API
`acf_zero_crossing_count`	G5	sign changes of the ACF	count	API
`acf_period_s`	G5	median of 2 × (spacing of ACF zero crossings) × Ts	s	EXP, CALC (CPLM-030), INV, REP, WB
`acf_regularity`	G5	min(1, P / (3·σ(periods)))	0–1	EXP, CALC (CPLM-031), REP, WB
`gate5_status`	G5	WARN when any ACF period exists, else PASS	enum	EXP strip, WB
`fft_peak_period_s`, freq, amplitude	G6	largest in-band Welch PSD peak; period = 1/f	s, Hz	API, WB
`fft_peak_ratio`	G6	peak-lobe energy / total energy	0–1	WB
`fft_peak_to_median`	G6	A_peak / median in-band amplitude (≥ 3 validates an FFT period)	ratio	WB (Why G6)
`harmonic_amplitude_ratio`	G6	(A2 + A3 + A5) / A1	ratio	EXP, CALC (CPLM-040), REP
`harmonic_energy_ratio`	G6	$(A2^2 + A3^2 + A5^2) / A1^2$	ratio	EXP, CALC (CPLM-041), HIST, REP
`spectral_entropy`	G6	normalised Shannon entropy of the spectrum	0–1	API
`gate6_status`	G6	WARN if peak_ratio > 0.5, else PASS	enum	EXP strip, WB
`period_status`, `validated_period_s`, `cycle_source`	G5/G6	ACF period in band and regular → VALID (ACF); else FFT period in band with peak/median ≥ 3 (FFT or harmonic fold k = 2–5); else NO_VALID_CYCLE	enum, s	INV ("no clean oscillation cycle")
`period_reject_reason`	G5/G6	PERIOD_OUT_OF_BAND or NO_VALID_CYCLE	enum	flags
`cycle_samples`	G7/G9	round(validated period / Ts)	count	API
`completed_cycles`, `valid_cycles`	G7/G9	whole cycles in the window; accepted (non-flat) cycles	count	WB (G7, G9 headlines)
`triangularity`	G7	mean per cycle of SSE_sine / (SSE_sine + SSE_triangle)	0–1	EXP, CALC (CPLM-050), INV, HIST, REP
`first_cycle_sse_sine`, `…_triangle`	G7	fit residuals of the first accepted cycle	—	API
`gate7_status`	G7	NOT_EVALUATED without a valid period · STRONG > 0.8 · else PASS	enum	EXP strip, WB
`horch_odd_sum`, `horch_even_sum`	G8	Σ|ϕ+ − ϕ−|, Σ|ϕ+ + ϕ−| over lags 1…200	—	API

Field	Gate	Formula / definition	Unit / range	On screen
horch_oddness	G8	odd / (odd + even)	0–1	EXP, CALC (CPLM-060), REP
gate8_status	G8	STRONG > 0.7, else PASS	enum	EXP strip, WB
phase_bbox_area	G9	(PVmax − PVmin) × (OPmax − OPmin)	PV·%	API
phase_area_norm_per_cycle / window_area_norm	G9	mean |shoelace area per cycle| / bbox area	0–1	WB (mean area)
phase_path_area	G9	Σ |cycle area|	PV·%	API
corner_score (raw)	G9	share of turning angle at corners ≥ class angle	0–1	EXP, CALC (CPLM-070), REP
corner_score_qualified	G9	same, only corners with an OP move ≥ deadband	0–1	fusion (geometry score)
gate9_status / reason	G9	NOT_EVALUATED (no cycles / area above band / effort or travel below floor) · PASS (area below band) · STRONG area > 0.30 · else PASS	enum	EXP strip, WB
limit_low_pp, limit_high_pp, limit_source	G10	5 pp and 95 pp of the declared (or 0–100) OP range	%	WB
saturation_pct (long)	G10	count(OP at a limit) / n	0–1	WB (Why G10)
sat_limit_dwell_samples, sat_limit_dwell_s	G10	longest run at a limit (samples, seconds)	count, s	WB (Why G10)
dwell_events[]	G10	each contiguous same-side run: start, end, duration, side	s	WB (dwell events)
sat_cycling_pattern	G10	≥ 3 exits and ≥ 2 % occupancy	bool	API
limit_dwell_warn_s	G10	configured dwell, else 30 samples × Ts	s	WB
gate10_valve_output	G10	WARN occupancy ≥ 0.05 or dwell ≥ warn or cycling, else PASS	enum	EXP strip, WB
vp_available, vp_eligible_samples	G10/G14	any VP sample; GOOD and fresh (≤ 60 s) VP samples	bool, count	readiness, flags (NO_VP)
vp_tracking_error_mean_pp, …_p95_pp	G10/G14	mean and P95 of |OP − VP|	pp	INV (G14 prose)
vp_nonfollowing_dwell_s	G14	longest run with |OP − VP| ≥ 5 pp	s	INV (G14 prose), WB
vp_tracking_healthy	G14	≥ 30 eligible samples ∧ P95 < 5 pp ∧ dwell < 60 s	bool	G14 CONTRADICTED
freeze_run_samples, freeze_index_s	G11	longest run of unchanged PV; × Ts	count, s	INV (freeze index)
freeze_fraction	G11	freeze_run_samples / n	0–1 (shown %)	EXP
freeze_material	G11	freeze ≥ 60 s ∧ ≥ 10 % of the window (→ EXCLUDED_SENSOR)	bool	diagnosis
freeze_events[]	G11	runs within tolerance lasting ≥ 60 s	s	WB (freeze events)
pv_quantization_count	G11	distinct round(PV × 1000) values	count	WB
pv_drift_per_day	G11	least-squares PV slope × 86 400	PV/day	API
delta_pv_mean, delta_pv_std, spike_count	G11	ΔPV statistics; spikes = |ΔPV − mean| > 3σ	PV, count	API
gate11_status / reason	G11	WARN MATERIAL_FREEZE or FREEZE_EPISODE, else PASS	enum	EXP strip, WB
interaction.peers[]	G13	per peer: support_s, peak_xcorr, lag_s, coherence_band_max, associated, direction	s, −1…1, 0–1	WB (G13 headlines)
interaction.state	G13	NOT_CONFIGURED / UNAVAILABLE / NO_INTERACTION / INTERACTION_DETECTED	enum	EXP strip (G13), WB
effort_ratio, op_range_pct (long)	G4/G9	σ(OP)/σ(PV); max OP − min OP	ratio, %	EXP, INV
travel_per_day, reversals_per_hour (long)	G4	as short, over the long window	%/day, 1/h	EXP, CALC (long rows)
long_metrics_qualified	—	false when the aligned short G0 failed	bool	WI (UNQUALIFIED)